

Exploring the Unique Extracellular Matrix Composition of *Acomys* as a Potential Key to Resisting Fibrosis

Michele N. Dill, Zoe Turner, Paulina W. Kapuscinska, Katie Heiden, Kari B. Basso, Chelsey S. Simmons, and Erika Moore*



Cite This: *ACS Biomater. Sci. Eng.* 2025, 11, 3616–3633



Read Online

ACCESS |



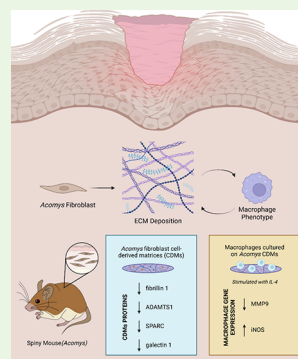
Metrics & More



Article Recommendations

ABSTRACT: Fibrosis is a dysregulated wound healing response characterized by excessive accumulation of dense scar tissue that inhibits organ function and is estimated to contribute to up to 45% of deaths in the industrialized world. In this work, we sought to uncover new ways to address fibrosis by drawing inspiration from an animal that does not develop fibrosis. The Spiny Mouse (*Acomys*) has the most extensive regenerative capabilities of any known mammal and can regenerate injuries to the skin, kidney, heart, skeletal muscle, and spine with little to no fibrosis. We hypothesize that the regenerative abilities of *Acomys* are due, in part, to altered stiffness-mediated fibroblast-to-myofibroblast transition (FMT). In this work, we interrogated stiffness-mediated FMT in *Acomys* and *Mus* dermal fibroblasts *in vitro* by performing RNA Sequencing and found no differential gene expression in *Acomys* fibroblasts cultured on soft vs stiff substrates. We further investigated the direct impact of stiffness-mediated FMT and species differences on ECM deposition by fabricating cell-derived matrices (CDMs) from *Acomys* and *Mus* fibroblasts cultured on varying stiffnesses. After assessing the composition of these CDMs using label-free quantitative proteomics, fibrosis-associated extracellular matrix proteins including fibrillin-1, ADAMTS1, SPARC, and galectin-1 were found to be significantly reduced or absent in *Acomys* CDMs compared to *Mus* CDMs. In addition, proteins that have been connected to fibrosis resolution, including Col12a1 and clusterin, were upregulated in *Acomys* CDMs. When cultured on *Acomys* CDMs, mouse macrophages downregulated MMP9 mRNA expression and maintained increased expression of iNOS in response to IL-4, a pro-fibrotic cytokine. These results indicate a direct impact of species-specific ECM compositions on macrophage phenotype and suggest that ECM produced by *Acomys* fibroblasts may impede the development of a pro-fibrotic macrophage phenotype in the presence of pro-fibrotic stimuli.

KEYWORDS: cell-derived matrices, fibroblast-to-myofibroblast transition, macrophages, stiffness-mediated signaling, fibrosis, spiny mouse (*Acomys*)



1. INTRODUCTION

Fibrosis is a dysregulated wound healing response characterized by the deposition of excess extracellular matrix (ECM) and accumulation of dense scar tissue around the injured site. It is estimated to contribute to 45% of deaths in the industrialized world and imposes a significant burden on the healthcare system globally.^{1,2} Chronic diseases such as idiopathic pulmonary fibrosis, liver cirrhosis, and heart failure are major contributors to this statistic. With this, there is a need for continuous research to build upon our current understanding of the mechanisms of fibrosis and inform treatment development. In this work, we sought to expand our understanding of fibrosis through exploring the unique regenerative capabilities of the Spiny Mouse (*Acomys*).

The Spiny Mouse (*Acomys*) has the most extensive regeneration of any known mammal and can regenerate full thickness skin wounds^{3–7} and burns,⁸ or damage to the ear pinna,^{9–11} skeletal muscle,¹² kidney,¹³ heart,^{14,15} and spinal cord^{16,17} with reduced or no fibrosis compared to standard laboratory mice. These abilities suggest an underlying

mechanism that may provide insight into preventing or reversing fibrotic processes in humans. We hypothesize that this regenerative ability in *Acomys* is due, in part, to reduced stiffness-mediated fibroblast-to-myofibroblast transition (FMT) during wound healing.

During normal wound healing, tissue resident fibroblasts differentiate into specialized myofibroblasts through the FMT to aid in restoring tissue integrity.^{18–20,20–23} Myofibroblasts are activated fibroblasts^{24,25} characterized by increased contraction, increased ECM production, and the de novo production of alpha smooth muscle actin (α SMA) colocalized to stress fibers.^{24–26} Transient myofibroblast activation is necessary for reestablishing tissue integrity through tissue

Received: April 25, 2025

Revised: May 7, 2025

Accepted: May 8, 2025

Published: May 23, 2025



contraction and the production of fibrillar, collagen-rich granulation tissue and can be resolved, resulting in restored tissue architecture.²⁷ However, persistent myofibroblast presence contributes to fibrosis pathogenesis through excessive ECM deposition and crosstalk with pro-fibrotic macrophages.^{25,28}

Stiffness-mediated FMT is a key mechanism that contributes to persistence of myofibroblasts and progression of fibrosis.^{29,30} When exposed to an environment stiffer than healthy tissue, fibroblasts differentiate into myofibroblasts.³¹ These myofibroblasts then contribute to tissue stiffening through increased deposition of ECM, creating a positive feedback loop that contributes to progressive fibrosis.^{26,29,32,33} Inhibiting stiffness sensing by fibroblasts has been shown to ameliorate fibrosis by reducing FMT and myofibroblast persistence. For example, inhibition of FAK signaling through pharmacological treatment or siRNA resulted in reduced α SMA expression and attenuated bleomycin-induced lung fibrosis in mice.³⁴ Similarly, deletion of the mechanosensors YAP/TAZ in mouse fibroblasts has been shown to prevent FMT and suppress fibrosis in a mouse model of renal fibrosis.³⁵ Blocking stiffness-mediated FMT through inhibition of TGF- β 1 activation reduced myofibroblast presence and fibrotic capsule thickness in response to a stiff PDMS implant.³⁰

Research suggests that reduced stiffness-mediated FMT may contribute to the regeneration of *Acomys* wounds. For example, regenerating *Acomys* ear pinna wounds demonstrate reduced presence and persistence of myofibroblasts *in vivo* compared to CD-1 mice (*Mus*).^{3,36} Similarly, reduced myofibroblast presence and fibrotic capsule formation was also observed following subcutaneous implantation of a stiff polydimethylsiloxane (PDMS) implant in *Acomys* compared to *Mus* (preprint).³⁷ *In vitro* studies further support this finding, as *Acomys* dermal fibroblasts do not undergo stiffness-mediated FMT when cultured on stiff PDMS or plastic.³⁸

The reduced myofibroblast activation observed in *Acomys* wounds likely contributes to the reduction in collagen-rich scar tissue observed during regeneration.^{3,4,7} This reduced scarring is accompanied by distinct differences in ECM composition observed between *Acomys* and *Mus* wounds. For example, *Acomys* wounds possess a higher collagen III/collagen I ratio and increased presence of fibronectin and tenascin C when compared to *Mus* wounds.^{3–5} These ECM differences may play a key role in promoting regeneration and influencing other cellular behaviors.

Specifically, differences in ECM composition between *Acomys* and *Mus* wounds may affect macrophage phenotype and behavior. Macrophages, known for their plasticity, adapt their function based on their microenvironment.³⁹ In wound healing and fibrosis, macrophages clear cellular debris through phagocytosis, secrete pro-inflammatory or pro-fibrotic cytokines, support proline biosynthesis and collagen production through production of arginase-1 (Arg-1),^{33,40} and regulate ECM turnover through the secretion of matrix metalloproteinases (MMPs). Different ECM compositions are known to differentially regulate macrophage phenotype.^{41–44} For example, solubilized ECM containing higher hyaluronic acid content promoted increased expression of M2-related cytokines, while ECM with degraded⁴² hyaluronic acid increased production of M1-related cytokines.⁴² In addition, macrophages seeded on decellularized ECM from older cardiac tissue had reduced expression of both pro- and anti-

inflammatory cytokines compared to decellularized ECM from younger cardiac tissue.⁴³

The impact of ECM composition on macrophage phenotype suggests that ECM deposition by *Acomys* fibroblasts in the wound environment may contribute to the phenotypic differences observed in macrophage populations between *Acomys* and *Mus* wounds. *Acomys* wounds demonstrate reduced infiltration of pro-inflammatory macrophages and lower production of inflammatory cytokines compared to scarring *Mus* wounds,^{7,11–13,45} However, a reduction in anti-inflammatory cytokines has also been documented in *Acomys* wounds. Brant et al. reported a reduction of IL-4 in *Acomys* wounds compared to scarring *Mus* wounds,⁴ which is associated with reduced fibrosis in other models.^{46–48} These findings suggest that macrophages in *Acomys* wounds may adopt an intermediary phenotype that is characterized by overall low cytokine production.

In this work, we sought to expand on functional assessments of *Acomys* FMT *in vitro*³⁸ by performing RNA sequencing on *Acomys* fibroblasts cultured on increasing substrate stiffness. We anticipated that an extensive gene-level characterization of the response of *Acomys* fibroblasts to stiffness may lead us to key gene targets responsible for the inhibited FMT observed in *Acomys* fibroblasts. With the understanding that fibrosis emerges from communication between fibroblasts, the ECM, and macrophages, we next sought to generate an *in vitro* model that allows for delineation of the role of ECM, secreted by fibroblasts, on macrophage phenotype. Specifically, we created cell derived matrices (CDMs), which are three-dimensional ECM scaffolds generated *in vitro* from stromal cells,⁴⁹ from *Acomys* and *Mus* fibroblasts on increasing substrate stiffnesses. By culturing *Acomys* and *Mus* fibroblasts on soft PDMS, stiff PDMS, and glass substrates and harvesting their resulting CDM, we sought to gain a better understanding of the role of stiffness in fibroblast ECM generation, as well as how this is altered between *Acomys* and *Mus*, a scarring counterpart. We subsequently cultured mouse bone marrow-derived macrophages (mBMDMs) on these decellularized *Acomys* and *Mus* CDMs to examine the impact of species-specific ECM compositions on macrophage phenotype under pro-inflammatory and anti-inflammatory conditions.

2. MATERIALS AND METHODS

2.1. Preparation of PDMS Substrates for RNA Sequencing.

Soft PDMS substrates were prepared by mixing the two components of the Sylgard 527 elastomer kit (Dow Corning) in a 1:1 ratio by weight and transferring to well plates to obtain approximately 1 mm thickness. The mixture was then degassed for an hour and cured overnight at 70 °C. Stiff PDMS substrates were prepared by mixing the two components of the Sylgard 182 elastomer kit (Dow Corning) in a 1:10 ratio by weight and then following the same procedure. Using these weight ratios, the Young's moduli of Sylgard 527 and Sylgard 182 are ~ 5 kPa and ~ 1 MPa, respectively.^{50,51} To prepare the substrates for cell culture, the PDMS surfaces were charged by plasma treatment using a Harrick Plasma Cleaner (20 s on High for Sylgard 527, 60 s for Sylgard 182). The surfaces were covered with sterile Milli-Q water and UV treated for 20 min to sterilize. Following sterilization, the Milli-Q water was removed and replaced with a sterile solution of 10 μ g/mL rat tail collagen I (Corning) in PBS and the substrates were incubated for 40 min at 37 °C. Tissue culture plastic (TCP) substrates were purchased sterile and pretreated with plasma gas. They were also coated by adding a 10 μ g/mL rat tail collagen I solution for 40 min at 37 °C. Following the 40 min incubation, the collagen solution was removed from all substrates,

replaced with sterile PBS, and the substrates were wrapped in parafilm and stored at 4 °C until use.

2.2. Isolation of Primary *Acomys* and *Mus* Fibroblasts. All animal procedures were conducted in accordance with the guidelines established by the United States Department of Agriculture (USDA) and the National Institutes of Health (NIH) for the care and use of laboratory animals. Experimental protocols were approved by the University of Florida Institutional Animal Care and Use Committee (IACUC). The use of *Acomys cahirinus* was approved under protocol number 202107707, and the use of *Mus musculus* (CD-1) was approved under protocol number 202103505. Animals were obtained from established breeding colonies maintained at the University of Florida.

Acomys pups were euthanized within 3 days of birth and the dorsal skin was removed. The tissue was incubated overnight in 0.2% Disperse II (Roche) in DMEM at 4 °C, after which the dermis and epidermis were separated with forceps. The dermis was dissociated by incubation in 0.24% collagenase type I (Gibco) in PBS at 37 °C for 60–90 min with constant agitation. A cell suspension was obtained using a vacuum filtered conical tube, and the cells were rinsed and cultured in DMEM/F12 supplemented with 10% fetal bovine serum (Gibco), 10% NuSerum (Corning), 1% Gentamicin/Amphotericin B (Gibco), and 0.1% insulin-transferrin-selenium (Gibco). Isolation of cells from CD-1 *Mus* pups followed the same procedure.

2.3. Cell Culture and RNA Isolation for RNA Sequencing. Immediately following isolation, primary fibroblasts from 3 male *Acomys* pups and 3 male *Mus* pups were seeded directly onto collagen coated soft PDMS, stiff PDMS, and TCP substrates at a density of 26,000 cells/cm². The fibroblasts were cultured for two additional passages, then RNA was isolated from 1 million cells per condition using QIAGEN's RNeasy Mini Kit. RNA Sequencing of 18 samples was performed by the UF ICBR Gene Expression & Genotyping Core using a poly A selection for library prep and an Illumina Nova Seq500. Analysis was performed by the UF ICBR Bioinformatics Core. Briefly, trimming and QC was performed using fastp (v 0.22.0) and reads were aligned to a recently published *Acomys cahirinus* genome⁵² using STAR (v 2.7.9a). Transcript abundance was quantified using RSEM (v1.3.1) and differential expression analysis was performed using DESeq2. Results were reported for protein-coding genes only. Cutoffs of FDR-corrected *p*-value ≤ 0.05 and |Log2FC| ≥ 1.0 were used to identify differentially expressed genes.

2.4. Preparation of Substrates for CDM Generation. For CDM substrate preparation, soft and stiff PDMS mixtures were made as described above and cured in well plates or onto glass coverslips. Functionalization of all substrates with collagen I was performed to promote cell adhesion in long-term culture. Briefly, substrates were plasma treated as described above and 1% (3-aminopropyl)-trimethoxysilane (APTMS) in 95% ethanol was immediately added to each well for 10 min to functionalize the substrates with free primary amine groups. Unreacted APTMS was removed by 3 rinses with sterile Milli-Q water, then sterile 1% glutaraldehyde in PBS was added and allowed to react for 30 min, leaving aldehyde groups. Residual glutaraldehyde was removed by 3 rinses with sterile PBS. A sterile solution of 10 µg/mL rat tail collagen I (Corning) was added to each well, and the well plates were incubated for 1 h at room temperature on a rocker. The collagen I solution was aspirated and replaced with sterile PBS, then the substrates were sterilized under a bactericidal UV lamp for 30 min. The plates were wrapped in parafilm and stored at 4 °C until use.

2.5. Generation of *Acomys* and *Mus* CDMs. Primary dermal fibroblasts from 3 *Acomys* and 3 *Mus* pups were isolated as previously described and immediately seeded onto soft PDMS coated with 10 µg/mL rat tail collagen I (Corning), expanded, and frozen for later use. For CDM generation, fibroblasts were seeded onto collagen functionalized PDMS or glass substrates at a density of 10,000 cells/cm² and cultured for 16–21 days with media changes every other day. Once cells reached confluency, the media was supplemented with 25 mg/mL Polysucrose 400 powder (Sigma-Aldrich), a commonly used macromolecular crowder that promotes ECM deposition by fibroblasts during CDM culture.^{53,54} CDM culture was stopped either

at 21 days, or when cell detachment began (16 days for some samples).

Decellularization of CDMs was performed using a prewarmed solution of 0.05% Triton X-100 (Sigma-Aldrich) and 20 mM NH₄OH in PBS for 5 min. CDMs were carefully rinsed 3 times with PBS, then treated with 10 U/mL DNaseI (Roche) for 30 min at 37 °C to remove residual DNA. Following treatment with DNaseI, the CDMs were rinsed 3 times with PBS and stored at 4 °C in PBS supplemented with 1% Amphotericin B and 1% PenStrep until further use.

2.6. Solubilization of CDMs for Label-Free Quantitative Proteomics. Decellularized CDMs were detached from the PDMS substrates by treating with RIPA buffer (Santa Cruz Biotechnology) for 5 min on ice, followed by scraping with a cell scraper. The CDM and RIPA buffer solution was transferred to a 15 mL conical tube and incubated for 30 min on ice. After 30 min, the CDMs were homogenized using a tissue homogenizer (Fisherbrand Hand-held Homogenizer). The samples were then centrifuged at 3200g and 4 °C for 15 min and the supernatant was removed and stored at –20 °C. The remaining pellet was resuspended in a membrane solubilization buffer containing 40 mM Tris-Cl (pH 8.0), 7 M urea, 2 M thiourea, 0.25% w/v ASB-14, and 0.25% NP-40 and incubated at room temperature for 15 min. The samples were vortexed, then centrifuged at 3200g for 30 min. Then, the supernatant was removed and diluted by 2.5 with distilled water and stored at –20 °C. Prior to analysis, both protein fractions were precipitated in ice-cold acetone overnight. Following precipitation, the samples were centrifuged at 3200g for 15 min. The acetone was removed through decanting and air drying, and the remaining protein pellets were resuspended in 2 M urea. The RIPA buffer soluble and urea soluble protein fractions from each sample were combined and the samples were delivered to the UF Mass Spectrometry Research and Education Center for label-free quantitative proteomics using liquid chromatography-tandem mass spectrometry.

2.7. Label-Free Quantitative Proteomics. Proteins were extracted and digested using the EasyPep MS Sample Prep Kit (Thermo Fisher Scientific). Total protein was determined on a Qubit and the appropriate volume of each sample was taken to equal 20 mg total protein for digestion. The samples were digested with the sequencing grade trypsin/lys C rapid digestion kit from Promega (Madison WI) using the manufacturer recommended protocol. Three times the sample volume of rapid digestion buffer (provided with the kit) was added to the samples. The sample was incubated at 56 °C with 1 mL of dithiothreitol (DTT) solution (0.1 M in 100 mM ammonium bicarbonate) for 30 min prior to the addition of 0.54 mL of 55 mM Iodoacetamide in 100 mM ammonium bicarbonate. Iodoacetamide was incubated at room temperature in the dark for 30 min. The trypsin/lys C was prepared fresh as 1 mg/mL in the rapid digestion buffer. One mL of enzyme was added, and the samples were incubated at 70 °C for 1 h. The digestion was stopped with the addition of 0.5% TFA. The MS analysis is immediately performed to ensure high quality tryptic peptides with minimal nonspecific cleavage.

Nanoliquid chromatography tandem mass spectrometry (Nano-LC/MS/MS) was performed on a Thermo Scientific Q Exactive HF Orbitrap mass spectrometer equipped with an EASY Spray nanospray source (Thermo Scientific) operated in positive ion mode. The LC system was an UltiMate 3000 RSLCnano system from Thermo Scientific. The mobile phase A was water containing 0.1% formic acid and the mobile phase B was acetonitrile with 0.1% formic acid. The mobile phase A for the loading pump was water containing 0.1% trifluoroacetic acid. 5 mL of sample was injected onto a Thermo Scientific mPAC C18 trapping column (C18, 5 mm pillar diameter, 10 mm length, 2.5 mm interpillar distance) at a 10 mL/min flow rate. This was held for 3 min and washed with 1%B to desalt and concentrate the peptides. The injector port was switched to inject, and the peptides were eluted off of the trap onto the column. A Thermo Scientific 110 cm mPAC was used for chromatographic separations (C18, 5 mm pillar diameter, 110 cm length, 2.5 mm interpillar distance). The column temperature was maintained at 40 °C. A flow rate of 750 nL/min was used for the first 15 min and then

the flow was reduced to 250 nL/min. Peptides were eluted directly off the column into the Q Exactive system using a gradient of 1% B to 20% B over 100 min and then to 45% B in 20 min for a total run time of 150 min (Table 1).

Table 1. LC Gradient Conditions of Time, Percent B, and Flow Rate

Time (min)	% B	Flow Rate (nL/min)
0	1	750
3	1	750
15	5	750
15.1	5	300
100	20	300
123	45	300
130	95	300
135	95	300
135.1	1	300
150	1	300

The MS/MS was acquired according to standard conditions established in the lab. The EASY Spray source operated with a spray voltage of 1.5 kV and a capillary temperature of 200 °C. The scan sequence of the mass spectrometer was based on the original TopTen method; the analysis was programmed for a full scan recorded between 375 – 1575 Da at 60,000 resolution and a MS/MS scan at resolution 15,000 to generate product ion spectra to determine amino acid sequence in consecutive instrument scans of the 15 most abundant peaks in the spectrum. The AGC Target ion number was set at 3e6 ions for full scan and 2e5 ions for MS² mode. Maximum ion injection time was set at 50 ms for full scan and 55 ms for MS² mode. The micro scan number was set at 1 for both full scan and MS² scan. The HCD fragmentation energy (N)CE/stepped NCE was set to 28 and an isolation window of 4 *m/z*. Singly charged ions were excluded from MS². Dynamic exclusion was enabled with a repeat count of 1 within 15 s and to exclude isotopes. A Siloxane background peak at 445.12003 was used as the internal lock mass.

HeLa protein digest standard was used to evaluate the integrity and the performance of the columns and mass spectrometer. If the number of protein IDs from the HeLa standard falls below 3200, the instrument is cleaned, and new columns are installed.

All MS/MS spectra were analyzed using the Chimerys node (Thermo Fisher Scientific, San Jose, CA, USA; Proteome Discoverer 3.0.1.27). Chimerys was set up to search *Mus musculus* (sp_incl_isoforms TaxID = 10090_and_subtaxonomies) (v2023-06-28), *Acomys cahirinus* (v2024-01-31 with 31141 sequences and 20612094 residues) and Universal Protein Contaminants fasta assuming the digestion enzyme trypsin. Chimerys was searched with a fragment ion mass tolerance of 0.020 Da and a precursor ion tolerance of 10.0 ppm. Carbamidomethyl of cysteine was specified in Chimerys as a fixed modification.

Precursor ion intensity label free quantitation was done using Proteome Discoverer (Thermo Fisher Scientific vs 3.01.1.27). The two groups in each comparison were compared using a “non-nested” study factor. Normalization was derived by using all peptides. Protein abundances were calculated by summed abundances, meaning the protein abundances are calculated by summing sample abundances of the connected peptide groups. Fisher’s exact test (pairwise ratio-based) was used to calculate *p*-values with low intensity resampling value imputation included. Adjusted *p*-values were calculated using Benjamini–Hochberg. Fold-change comparisons and identification of unique protein compositions were performed with Scaffold DDA (version 6.3.0) from Proteome Software.

2.8. Macrophage Culture on Decellularized CDMs. C57/BL6 mouse bone marrow-derived macrophages (mBMDMs) (Cell Biologics) were thawed and seeded directly onto *Acomys* and *Mus* CDMs at a density of 150,000 cells/cm². The cells were cultured for 72 h in macrophage media supplemented with 10% FBS and 1% PenStrep

(Cell Biologics) to allow attachment to CDMs and recovery from cryopreservation. After 72 h, the media was replaced with fresh macrophage media, macrophage media containing 100 ng/mL LPS and 10 ng/mL IFN γ , or macrophage media containing 20 ng/mL IL-4, and the macrophages were cultured for an additional 48 h. These treatments were intended to polarize macrophages toward an M0-, M1-, and M2-like phenotype, respectively.

2.9. RNA Isolation and Gene Expression Analysis with Nanostring nCounter. Following macrophage seeding and polarization, CDMs were degraded using 0.25% Trypsin-EDTA for 5–7 min at 37 °C. The mBMDMs and degraded CDMs were transferred into 15 mL conical tubes and centrifuged at 200g for 5 min. The cell pellet was rinsed in PBS and centrifuged again at 200g for 5 min. RNA was isolated using the Qiagen RNeasy Mini Kit, and concentration and quality were assessed with a Nanodrop. The RNA samples were hybridized to the Nanostring nCounter mouse Fibrosis_V2 Panel barcoded Reporter CodeSet and Capture ProbeSet following the manufacturer protocol. Following hybridization, samples were immediately frozen and sent to the NanoString CX Lab to be run on a NanoString SPRINT Profiler. Raw data was returned and analyzed with nSolver Analysis Software (version 4.0). Significance was determined by a Benjamini–Yekutieli FDR-adjusted *p*-value <0.5 and lLog2FCI > 0.6.

2.10. Immunofluorescence Staining and Quantification of iNOS and Arg-1. Following macrophage seeding and polarization, samples were fixed with 4% PFA and stored in PBS at 4 °C until staining. Prior to staining, the fixed cells were permeabilized with 0.01% Triton X-100 for 10 min, then the samples were rinsed 3 times in PBS. The samples were incubated in a 3% BSA blocking solution for 2 h. Then, the blocking solution was replaced with a 3% BSA solution containing rabbit anti-iNOS (1:100, Invitrogen, PA3-030A) and mouse anti-Iba1 (1:100, Proteintech, 66129-1-Ig), and the samples were incubated for 2 h at room temperature. Following the incubation in primary antibody solution, the samples were rinsed with PBS for 1 h at room temperature. The samples were then incubated in a secondary antibody solution of a 3% BSA solution containing Alexa Fluor 555 donkey antirabbit IgG (1:400, Invitrogen A31572) and Alexa Fluor 488 donkey antimouse IgG (1:400 Invitrogen A21202) for 40 min at room temperature. Following incubation with the secondary antibody solution, the samples were rinsed with PBS overnight at 4 °C. The samples were stained with DAPI (1.5 μ g/mL) for 10 min, then rinsed gently with PBS. Then the samples were mounted to rectangular coverslips using antifade mounting media (Vectashield H-1400, Vector Laboratories). Fluorescence images were acquired with a Keyence microscope (BZ-X810) using Plan Fluor 20X (NA 0.45) objective.

A FIJI macro was generated to quantify the number of nuclei and mean intensity of iNOS and Arg-1 per field of view. The mean intensity values of iNOS and Arg-1 were determined, and three fields of view were averaged for each sample. Four experimental replicates were assessed per condition, using CDMs generated from fibroblasts from two different animals (*N* = 2, *n* = 8).

2.11. Zymography and Analysis of the Conditioned Media.

Conditioned media was collected from mBMDMs cultured on *Acomys* or *Mus* CDMs 48 h after polarization and stored at –80 °C until analysis. Protein concentration was determined using a bicinchoninic acid (BCA) protein assay following the manufacturer’s instructions (Thermo Scientific). Absorbance was measured at 562 nm using a Spark plate reader (Tecan). The average absorbance of each sample and standard was calculated, and a linear regression was applied to the BSA standard curve in Excel. The resulting equation was used to interpolate protein concentrations of experimental samples.

Gelatin zymography was used to assess MMP9 activity in the conditioned media. A total of 20 μ g of protein per sample, as determined by the BCA assay, was mixed with nonreducing sample buffer and loaded onto 10% zymogram gelatin gels (Novex). A molecular weight protein ladder (Thermo Fisher Scientific) and 125 ng of recombinant mouse MMP9 protein (Abcam) were included on each gel as size and activity controls, respectively. Gels were run at 125 V and 0.03 mA for approximately 120 min at 4 °C in 1 $\times$ Tris-

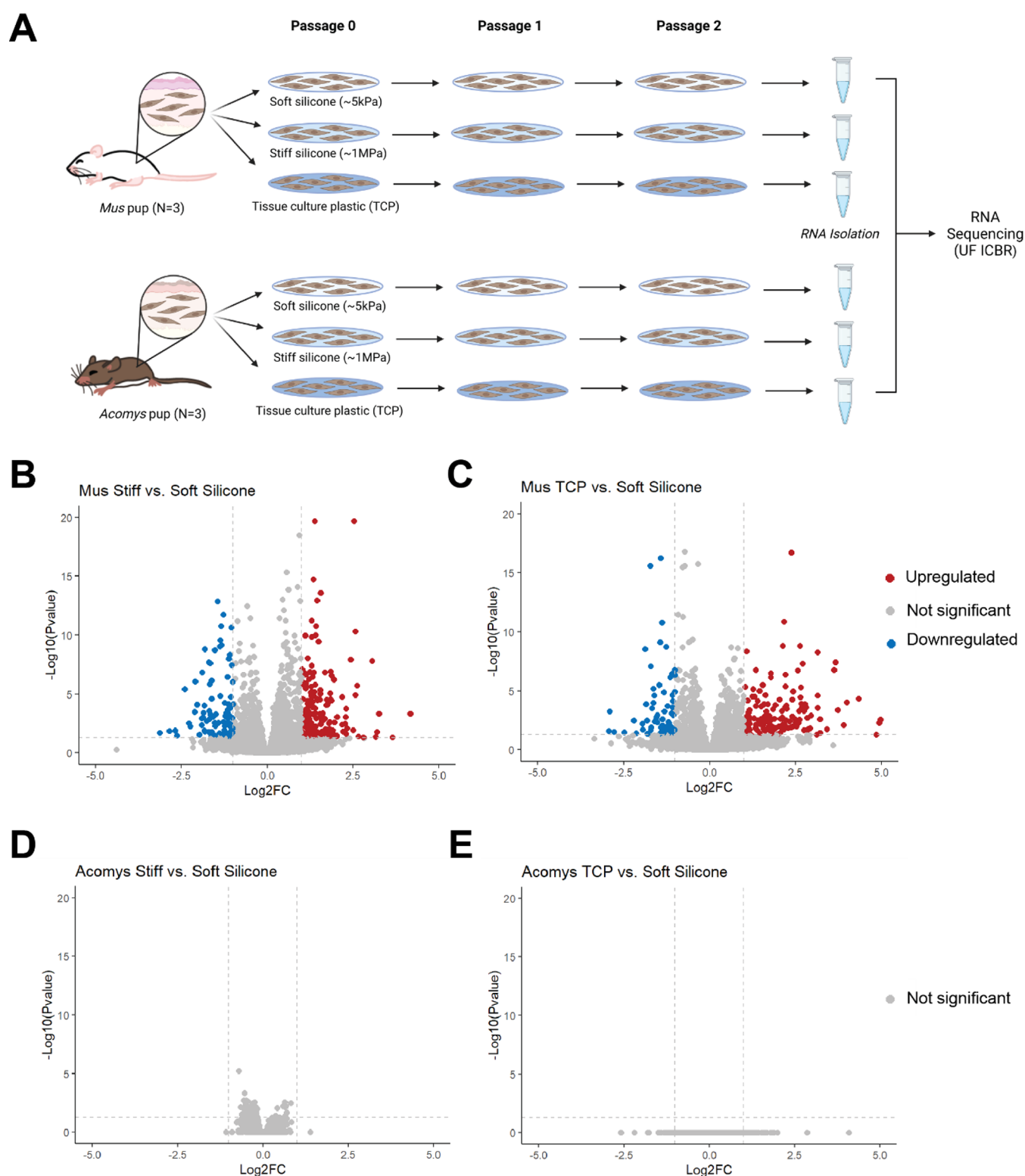


Figure 1. Substrate stiffness does not significantly alter gene expression in *Acomys* fibroblasts. (A) Experimental methods for RNA sequencing analysis of primary *Acomys* and *Mus* fibroblasts cultured on varying substrate stiffnesses. *Mus* fibroblasts cultured on (B) stiff PDMS and (C) tissue culture plastic (TCP) demonstrated differential gene expression compared to those cultured on soft PDMS. *Acomys* fibroblasts cultured on (D) stiff PDMS and (E) TCP did not demonstrate significant gene expression compared to those cultured on soft PDMS. Cutoffs for significance were $p < 0.05$ and $|\text{fold change}| > 2$.

Glycine SDS running buffer (Novex). Following electrophoresis, gels were incubated in 1× renaturation buffer (Novex), followed by 1× development buffer both at room temperature for 30 min each with gentle agitation to remove SDS and restore enzyme activity. Gels were

then transferred to 1× development buffer (Novex) and incubated at 37 °C overnight to allow substrate digestion.

After incubation, gels were washed twice in deionized water for 15 min each, stained for 1 h using Imperial Protein Stain (Thermo Fisher

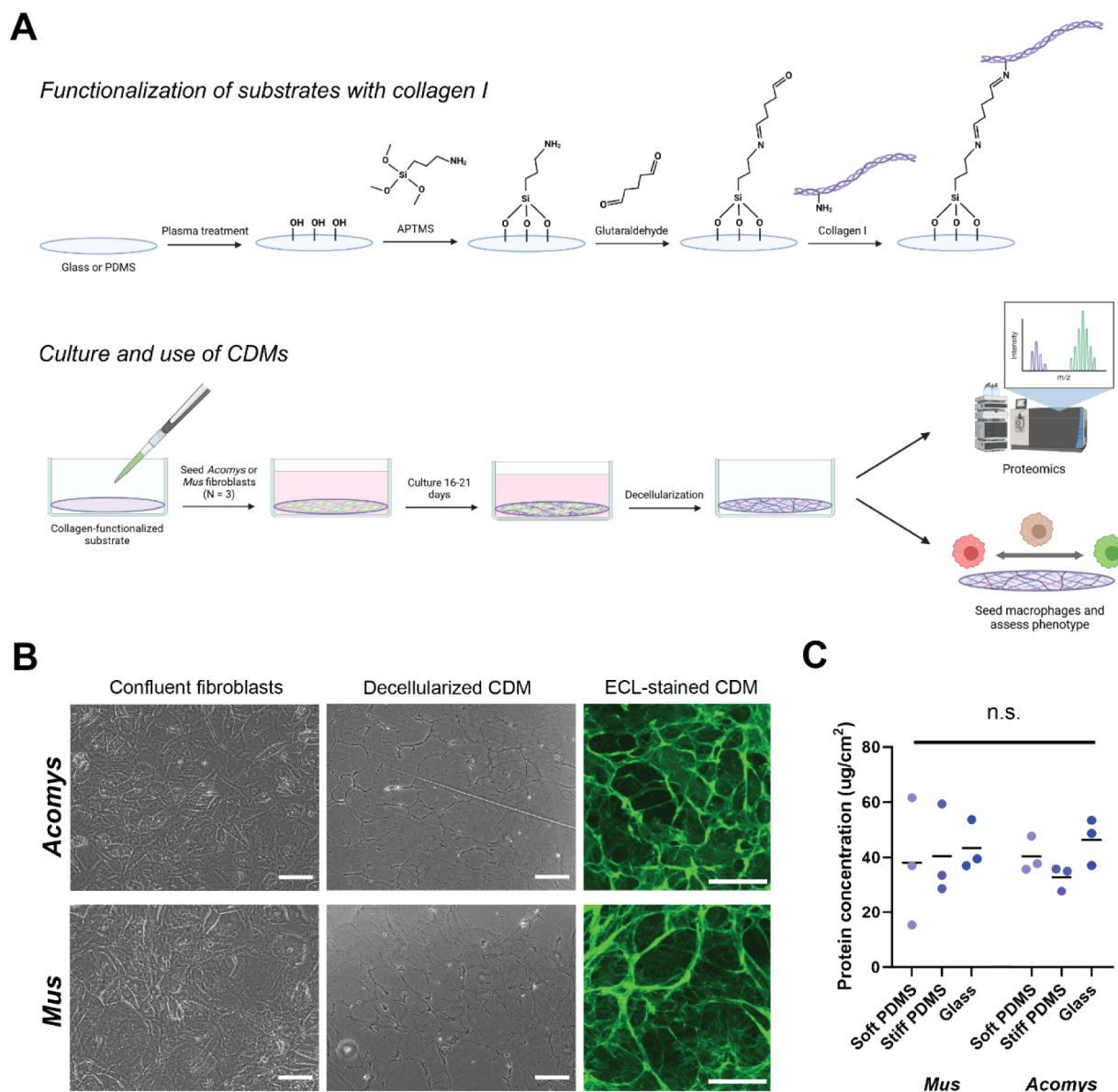


Figure 2. Fabrication and culture of CDMs from primary *Acomys* and *Mus* fibroblasts. (A) Procedure for collagen I functionalization of PDMS and glass substrates and culture of CDMs. (B) Representative images of *Acomys* and *Mus* CDMs in culture, following decellularization, and after staining with FITC conjugated *Erythrina crista-galli* Lectin (ECL), a nonspecific glycoprotein stain used for visualization of CDMs. Scale = 100 μ m. (C) Quantification of total protein per surface area obtained from each CDM group showed no significant difference in protein production across groups. Statistical significance between groups was assessed using a nonparametric Wilcoxon/Kruskal–Wallis test.

Scientific), and subsequently washed twice in deionized water for 30 min, until clear bands representing proteolytic activity were visible. Gels were imaged using LI-COR Odyssey CLx at 169 μ m resolution, medium scan quality, 0.5 intensity, and 0.5 mm focus offset range. Image analysis was performed in ImageJ. Lanes were defined and plotted using the gel analyzer tool, and the area under the curve (AUC) was measured for each unstained band to quantify relative MMP9 protease activity. AUC values were normalized to the recombinant mouse MMP9 control band intensity.

3. RESULTS

3.1. *Acomys* Fibroblasts Do Not Demonstrate Significant Differential Gene Expression in Response to Increased Substrate Stiffness. RNA Sequencing was performed using mRNA from primary *Acomys* ($N = 3$) and *Mus* ($N = 3$) fibroblasts that were cultured on soft PDMS (~ 5 kPa), stiff PDMS (~ 1 MPa), or TCP (~ 1 GPa) for 3 passages

to assess differential gene expression in response to substrate stiffness (Figure 1A). The fibroblasts were seeded on each substrate immediately upon isolation to avoid mechanical priming and were cultured for two additional passages to obtain sufficient cell numbers and stable phenotypes. Differentially expressed genes were analyzed within each species between fibroblast cultured on stiff PDMS vs soft PDMS and fibroblasts cultured on TCP vs soft PDMS. In all comparisons, the soft PDMS were used as the reference in Log2FC calculations, and cutoffs of FDR-adjusted p -value ≤ 0.05 and $|\text{Log2FC}| \geq 1$ were used. *Mus* fibroblasts cultured on stiff PDMS upregulated 142 genes and downregulated 69 genes compared to *Mus* fibroblasts cultured on soft PDMS (Figure 1B). Similarly, *Mus* fibroblasts cultured on TCP upregulated 186 genes and downregulated 87 genes compared to *Mus* fibroblasts cultured on soft PDMS (Figure 1C). Pathway

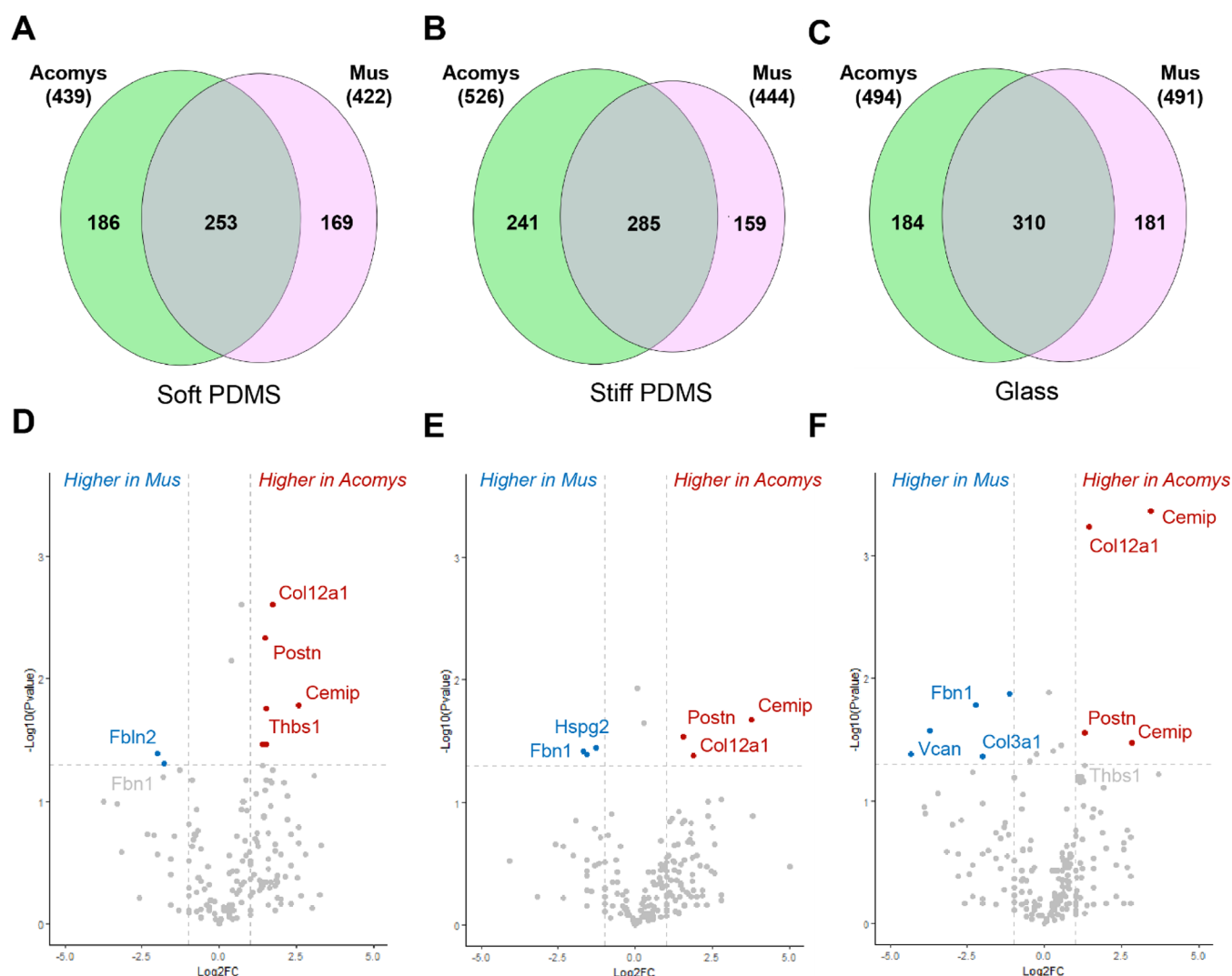


Figure 3. Differential protein expression between *Acomys* and *Mus* CDMs. Summary of shared and unique proteins in *Acomys* and *Mus* CDMs cultured on (A) soft PDMS, (B) stiff PDMS, or (C) glass. Differential expression of proteins present in both *Acomys* and *Mus* CDMs cultured on (D) soft PDMS, (E) stiff PDMS, and (F) Glass. ECM proteins upregulated in *Acomys* compared to *Mus* are labeled red while ECM proteins upregulated in *Mus* compared to *Acomys* are labeled blue. Cutoffs for significance were $p < 0.05$ and $|\text{fold change}| > 2$.

analysis of differentially expressed genes in *Mus* fibroblasts identified high overlap with pathways associated with degradation of the ECM, assembly of collagen fibrils, atherosclerosis signaling, hepatic fibrosis, and pulmonary fibrosis. In stark comparison, *Acomys* fibroblasts cultured on stiff PDMS or TCP did not significantly upregulate or downregulate any genes (Figure 1D,E). Samples from both species had similar quality control metrics for number of reads and transcriptome alignment rate, confirming that a lack of differentially expressed genes in *Acomys* was not due to technical issues.

3.2. Label-Free Quantitative Proteomic Analysis Identified Unique Protein Compositions between *Acomys* and *Mus* CDMs. To interrogate the deposition of ECM by primary *Acomys* and *Mus* fibroblasts exposed to stiff substrates for an extended period, CDMs were generated from low passage primary *Acomys* ($N = 3$) and *Mus* ($N = 3$) fibroblasts cultured on collagen-functionalized soft PDMS, stiff PDMS, or glass coverslips (Figure 2A,B). The substitution of TCP well plates with glass coverslips was made so the CDMs could be mounted on slides for imaging. The CDMs were

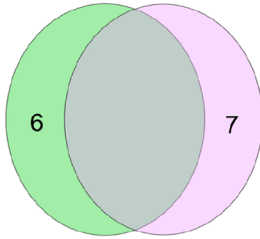
cultured for up to 21 days to mimic the beginning of fibrotic capsule development *in vivo*. Note that a longer culture time was not possible due to cell detachment in some samples after approximately 16 days. At the termination of the culture period, there were no significant differences in the total protein isolated from CDMs based on species ($p = 0.8252$) or stiffness ($p = 0.1276$), determined by a BCA assay (Figure 2C).

Label-free quantitative proteomic analysis of decellularized *Acomys* and *Mus* CDMs identified 253, 285, and 310 proteins shared between *Acomys* and *Mus* CDMs on soft PDMS, stiff PDMS, and glass, respectively (Figure 3A–C). Quantification of differential expression among these shared proteins, however, revealed significant upregulation of the ECM-related proteins cell migration-inducing hyaluronan-binding protein (CEMIP), collagen XII (COL12A1), periostin, and thrombospondin-1 in *Acomys* CDMs on soft PDMS while fibulin-2 was significantly upregulated in *Mus* CDMs (Figure 3D). Periostin, CEMIP, and COL12A1 were also upregulated in *Acomys* CDMs cultured on stiff PDMS and glass substrates (Figure 3E,F). When we quantified differential expression of proteins across *Acomys* CDMs cultured on different stiffnesses, we saw no

A

Proteins present in only *Acomys* CDMs

SC	Protein	Function (UniProt)
22	Vwa1	ECM assembly
15	Adamtsl6	FBN1 matrix assembly, attenuates TGFβ signaling
13	Smpdl3b	Negative regulator of TLR signaling
8	Loxl4	Stabilization of collagen and elastin fibers
29	Itih5	Tumor suppressor, hyaluronan metabolism
17	Clu	Cell proliferation, differentiation, apoptosis



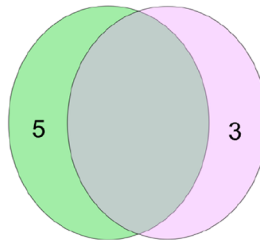
Proteins present only in *Mus* CDMs

SC	Protein	Function (UniProt)
15	Tnxb	Collagen fibril formation, inhibits cell migration
13	Matn2	ECM assembly
36	Adamts1	Proteolysis, inflammation, inhibits angiogenesis
10	Adamtsl4	FBN1 microfibril formation, apoptosis
17	Lgals3bp	Cell adhesion
14	Aspn	Calcium binding, negative regulation of TGFβ receptor signaling
7	Angptl2	Cell-cell signaling, endothelial cell sprouting

B

Proteins present only in stiff *Acomys* CDMs

SC	Protein	Function (UniProt)
6	Ltbp2	Elastic fiber organization/assembly
17	PCPE-1	Enhances procollagen C-proteinase activity
5	Pxdn	Collagen IV, fibronectin/laminin interactions, angiogenesis
9	Serpine1	Cell migration, cell response to IL-1, LPS, TGFβ
5	Tgfb2	Growth factor signaling



Proteins present only in stiff *Mus* CDMs

SC	Protein	Function (UniProt)
7	SPARC	Response to growth factors, binds to ECM proteins including collagens
4	Lgals1	Apoptosis, proliferation, differentiation, inflammatory response
6	Anxa5	Anticoagulant, apoptosis

Structural proteins

Proteases/enzymes

Other secreted proteins

Figure 4. Unique proteins in *Acomys* and *Mus* CDMs. (A) List of secreted proteins unique to *Acomys* or to *Mus* CDMs cultured on soft PDMS, stiff PDMS, and glass. (B) List of secreted proteins unique to *Acomys* or to *Mus* CDMs cultured on stiff PDMS and glass. SC denotes spectral counts, or the number of spectra identified for each protein.

significant differences with stiffness. In comparison, stiffness had an impact on protein expression in *Mus* CDMs, which had increased fibrillin-1, perlecan, versican, and collagen III (COL3A1) when cultured on stiff substrates (Figure 3E,F).

In addition to proteins shared between the two species, there were proteins identified only in *Acomys* or only in *Mus* CDMs in each comparison (Figure 4A,B). Specifically, there were 186 proteins unique to *Acomys* CDMs and 169 proteins unique to *Mus* CDMs on soft PDMS. On stiff PDMS, there were 241 proteins unique to *Acomys* and 159 unique to *Mus*. On glass, there were 184 proteins unique to *Acomys* and 181 proteins unique to *Mus*. To identify ECM-related proteins that were unique in each species, all unique proteins were filtered for secreted proteins and ≥ 2 spectral counts, and proteins that were shared between all *Acomys* or all *Mus* conditions were selected. We identified 6 proteins that were present in all *Acomys* samples and 7 proteins that were identified in all *Mus* samples but were not shared between species (Figure 4A). Proteins unique to *Acomys* CDMs included von Willebrand factor A domain containing 1 (VWA1), a disintegrin and metalloproteinase with thrombospondin motif-like 6 (ADAMTSL6), sphingomyelin phosphodiesterase acid like 3B (SMPDL3B), lysyl oxidase like 4 (LOXL4), interalpha-trypsin inhibitor heavy chain 5 (ITIHS), and clusterin. Proteins unique to *Mus* CDMs included tenascin-X, matrilin-2, a disintegrin and metalloproteinase with thrombospondin type motif 1 (ADAMTS1), a disintegrin and metalloproteinase with thrombospondin type motif-like 4 (ADAMTSL4),

galectin binding protein 3 (LGALS3BP), asporin, and angiopoietin-like protein 2 (ANGPTL2). To identify unique proteins mediated by increased stiffness, we then identified proteins that were present in all stiff (stiff PDMS and glass) samples in *Acomys* and all stiff samples in *Mus* (Figure 4B). Latent transforming growth factor beta binding protein 2 (LTBP2), procollagen C-endopeptidase enhancer (PCPE-1), peroxidase, serpin-1, and TGF- β 2 were unique to *Acomys* CDMs cultured on stiff substrates. Secreted protein acidic and rich in cysteine (SPARC), galectin-1, and annexin A5 were unique to *Mus* CDMs cultured on stiff substrates.

3.3. Mouse Macrophages Cultured on *Acomys* and *Mus* CDMs Have Modest Differences in Phenotype. To assess potential effects of compositional differences in *Acomys* and *Mus* CDMs on macrophage phenotype, mouse bone marrow-derived macrophages (mBMDMs) were cultured on decellularized *Acomys* ($N = 3$) or *Mus* ($N = 3$) CDMs under standard culture conditions, or with M1 or M2 stimulation treatments (Figure 5A). M1 stimulation consisted of a 48-h treatment with 100 ng/mL LPS and 10 ng/mL IFN γ , while M2 stimulation was a 48-h treatment 20 ng/mL IL-4. After stimulation treatments, RNA was isolated from the mBMDMs, and expression of fibrosis-related genes were assessed using the Nanostring nCounter Analysis System.

We first quantified differential gene expression between species by pooling all samples that passed quality control cutoffs. We found that mBMDMs cultured on *Acomys* CDMs downregulated MMP9 1.78-fold (FDR-adjusted $p = 0.0102$)

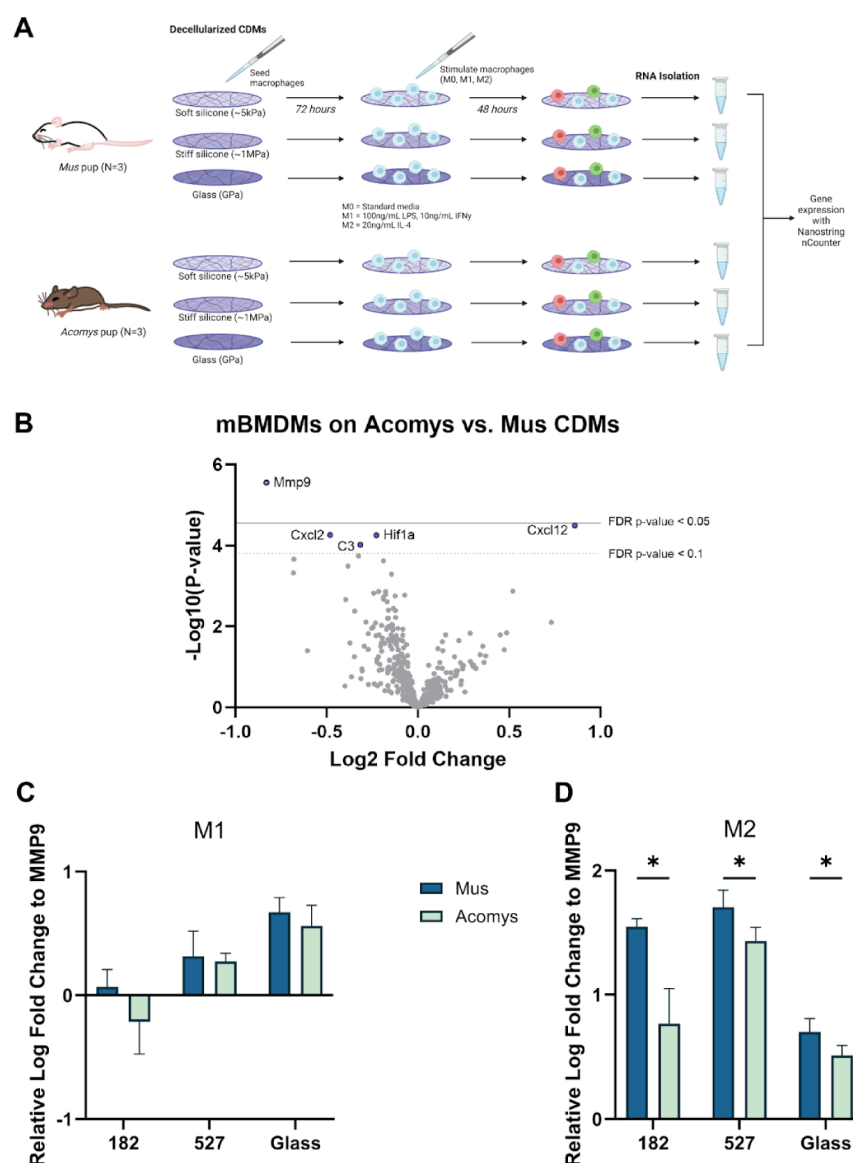


Figure 5. Mouse bone marrow-derived macrophages cultured on *Acomys* CDMs significantly downregulate MMP9 compared to those cultured on *Mus* CDMs. (A) Experimental setup for mBMDM culture on CDMs, stimulation, and isolation for NanoString nCounter gene expression analysis. (B) Volcano plot of differential gene expression in mBMDMs cultured on *Acomys* CDMs vs *Mus* CDMs with all samples pooled. Significance was determined by a Benjamini–Yekutieli FDR-adjusted p -value < 0.05 . Genes that were not significantly differentially expressed are indicated in gray. Measured Area under the curve (AUC) of MMP9 activity in conditioned media from mBMDMs cultured on *Mus* or *Acomys* CDMs following (C) M1 or (D) M2 polarization. AUC values shown are relative to the loaded recombinant mouse MMP9 protein. Significance was determined by a two-way ANOVA with Tukey's multiple comparisons test, $*p < 0.05$.

compared to mBMDMs cultured on *Mus* CDMs (Figure 5B). Using an FDR-adjusted p -value of 0.05, there were no other genes significantly expressed between mBMDMs cultured on *Acomys* compared to *Mus* CDMs.

We further investigated this downregulation of MMP9 from mBMDMs cultured on *Acomys* CDMs by running zymography on the conditioned media collected 48 h after polarization. A three-way ANOVA revealed that mBMDM polarization had the largest effect on MMP9 activity ($F(2, 27) = 44.26$, $p = 3.00 \times 10^{-9}$). Both CDM species ($F(1, 27) = 22.74$, $p = 5.67 \times 10^{-5}$), and substrate stiffness ($F(2, 27) = 21.76$, $p = 2.35 \times 10^{-6}$) also had statistically significant effects on MMP9 activity, but their impact was comparatively smaller based on the relative F -statistics and associated degrees of freedom. In addition, there was a significant three-way interaction between species,

polarization, and stiffness ($F(2, 27) = 4.40$, $p < 0.05$), indicating that the combined effects of polarization and stiffness on MMP9 activity were not consistent across *Mus* and *Acomys*. Based on this observation, we focused on M1 and M2 polarized mBMDMs (Figure 5C,D).

A two-way ANOVA for the M1-polarized conditioned media revealed only stiffness ($F(2, 14) = 14.00$, $p < 0.001$) significantly influenced MMP9 activity (Figure 5C). In contrast, the two-way ANOVA for the M2 mBMDMs showed significant effects of both stiffness ($F(2, 13) = 15.78$, $p < 0.001$) and species ($F(1, 14) = 13.09$, $p < 0.005$) on MMP9 activity (Figure 5D). Tukey's multiple comparisons test further revealed that across all stiffness conditions, M2 polarized mBMDMs cultured on *Acomys* CDMs exhibited significantly

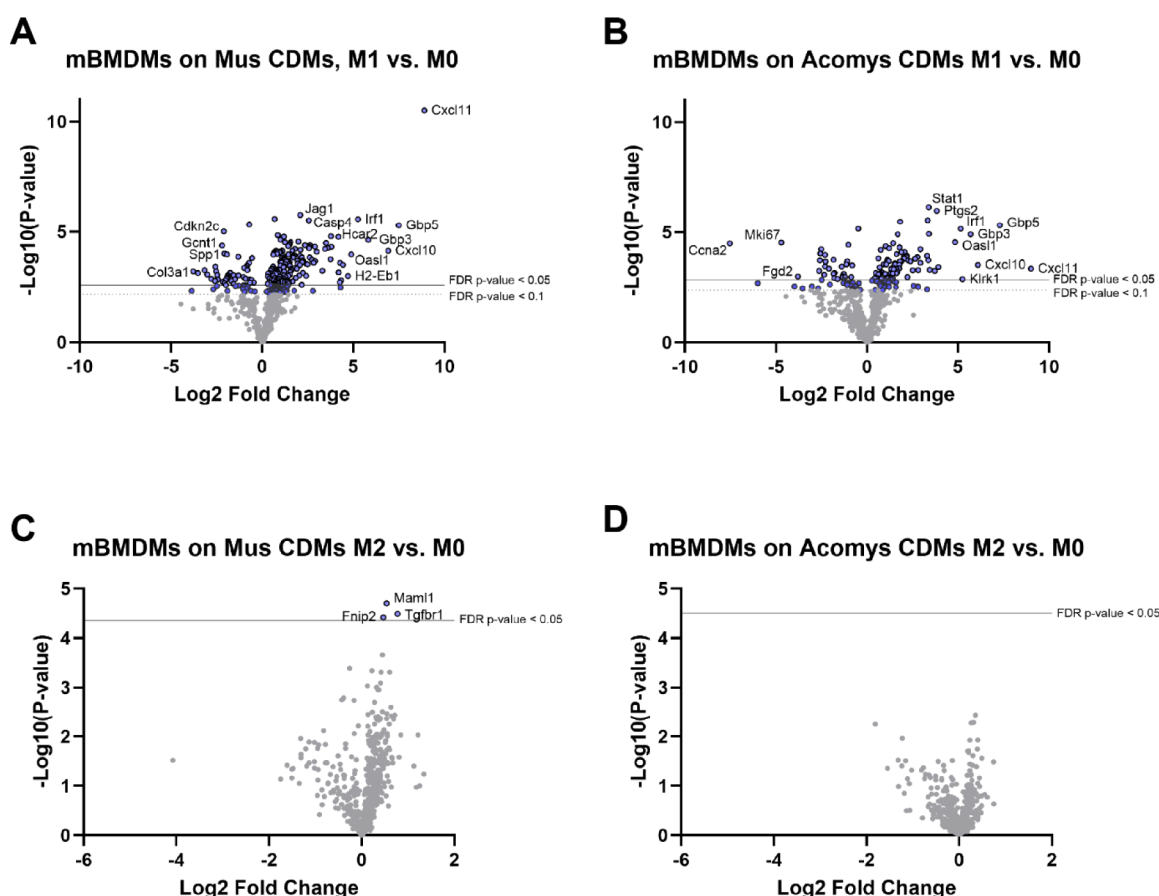


Figure 6. MBMDMs cultured on *Acomys* CDMs have a reduced overall response to M1 and M2 stimulation. Volcano plots of differential gene expression in mBMDMs cultured on (A) *Mus* and (B) *Acomys* CDMs in response to M1 stimulation or on (C) *Mus* and (D) *Acomys* CDMs in response to M2 stimulation. Significance was determined by a Benjamini–Yekutieli FDR-adjusted p -value < 0.5. Genes that were not significantly differentially expressed are indicated in gray.

downregulated MMP9 activity compared to those on *Mus* CDMs ($*p < 0.05$).

These results support the gene expression finding that MMP9 is downregulated in mBMDMs cultured on *Acomys* CDMs, as CDM species had a statistically significant effect on MMP9 activity identified by the three-way ANOVA. Particularly, M2 polarized mBMDMs demonstrated that stiffness and species influence MMP9 activity. Together, these data demonstrate that *Acomys* CDMs downregulate MMP9 at both the transcriptional and enzymatic levels.

Next, we quantified differential expression within mBMDMs seeded on *Acomys* or *Mus* CDMs generated on stiff PDMS in response to M1 or M2 stimulation. The purpose of this comparison was to assess the effects of ECM composition on mBMDMs in a pro-inflammatory or pro-fibrotic environment. In each comparison, the M0 treatment is used as the baseline. Note that the M0 baseline treatments were also from soft PDMS samples due to loss of one stiff PDMS sample from each species. When cultured on *Mus* CDMs, mBMDMs differentially expressed 378 genes in response to M1 treatment (Figure 6A). In comparison, when cultured on *Acomys* CDMs, mBMDMs differentially expressed fewer genes (234 genes) (Figure 6B). In comparison, the M2 treatments had a minimal effect on differential expression. In response to M2 treatment, mBMDMs cultured on *Mus* CDMs upregulated 3 genes (Figure 6C). MBMDMs cultured on *Acomys* CDMs did not differentially express any genes in response to M2 treatment

(Figure 6D). Further comparisons between each treatment group were limited due to unexpected loss of samples during hybridization, potentially due to inconsistencies in the thermocycler or failed seals in the PCR tubes.

Protein-level expression of Arg-1 and iNOS in mBMDMs cultured on *Acomys* and *Mus* CDMs was assessed through immunofluorescence staining (Figure 7A). Arg-1 and iNOS mean fluorescence intensities were quantified under three macrophage activation states: M0 (unstimulated), M1 (pro-inflammatory state induced by IFN- γ and LPS treatment), or M2 (anti-inflammatory state induced by IL-4 treatment). In this comparison, all CDMs were cultured on glass.

Expression of Arg-1, a marker associated with an M2 macrophage phenotype, in mBMDMs was not significantly affected by CDM species ($p = 0.645$) or stimulation ($p = 0.217$) (Figure 7B). However, expression of iNOS, a marker of the M1 phenotype, was significantly influenced by both species ($p = 0.00003$) and stimulation ($p < 0.00001$). Specifically, iNOS expression was significantly higher in mBMDMs cultured on *Acomys* CDMs compared to those cultured on *Mus* CDMs in the presence of IL-4 ($20.3 \text{ MFI} \pm 1.02$ and $12.68 \text{ MFI} \pm 0.73$, $p = 0.001$). These results suggest that the composition of *Acomys* CDMs support a macrophage phenotype with increased iNOS expression, even under M2 stimulating conditions.

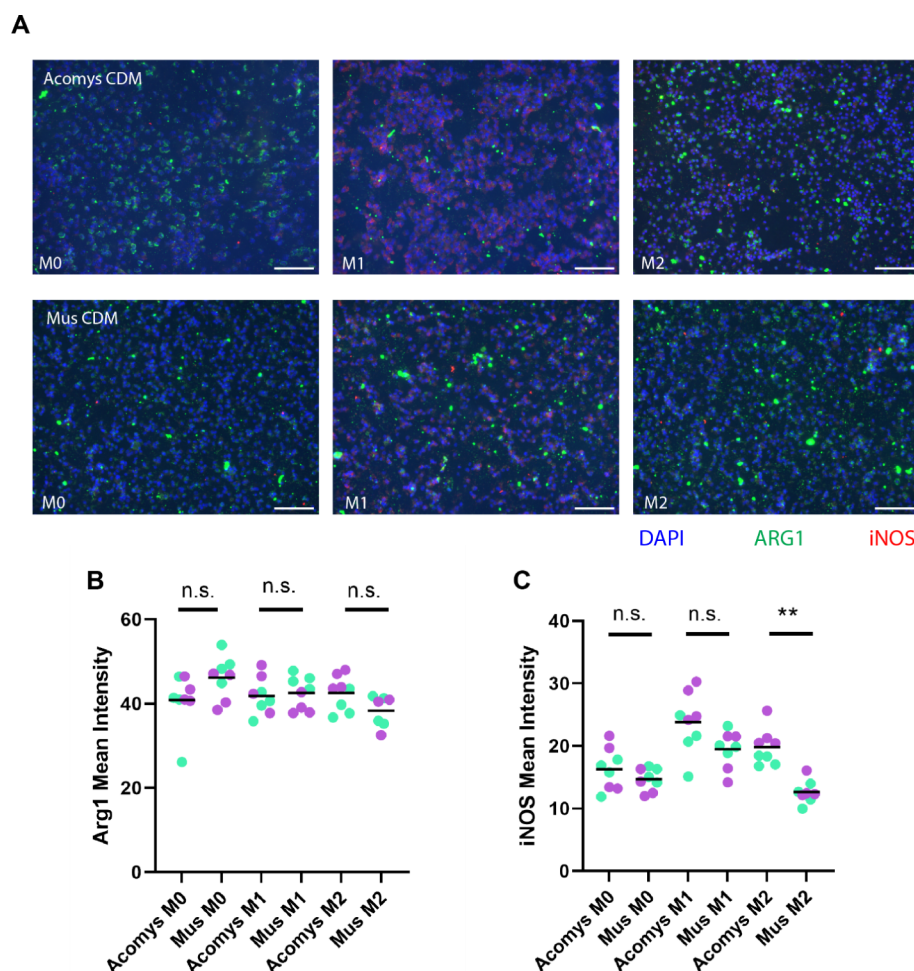


Figure 7. mBMDMs cultured on *Acomys* CDMs express increased iNOS under IL-4 treatment compared to mBMDMs cultured on *Mus* CDMs. (A) Immunofluorescence images of mBMDMs cultured on *Acomys* (top) or *Mus* (bottom) CDMs grown on glass coverslips and stimulated toward an M0, M1, or M2 phenotype. Scale bar = 100 μ m (B) Quantification of Arg-1 mean fluorescence intensity, where each point is the average of 3 fields of view per CDM. Colors indicate two different biological replicates. (C) Quantification of iNOS mean fluorescence intensity. Significance was determined by a two-way ANOVA with a Tukey's HSD post hoc test.

4. DISCUSSION

RNA Sequencing of *Acomys* and *Mus* fibroblast cultured on soft PDMS, stiff PDMS, or TCP revealed that *Acomys* fibroblasts did not significantly alter gene expression in response to substrate stiffness. In contrast, stiffness had a significant impact on differential gene expression in *Mus* fibroblasts. These results build upon functional assessments of stiffness-mediated FMT in primary *Acomys* and *Mus* fibroblasts³⁸ and confirm that stiffness-mediated FMT is inhibited in *Acomys* dermal fibroblasts. Inhibition of fibroblast mechanosensing has been shown to attenuate fibrosis in mouse models of pulmonary fibrosis,^{34,55} renal fibrosis,³⁵ and scleroderma.⁵⁶ Uncovering mechanisms behind reduced stiffness-mediated FMT in *Acomys* may lead to therapeutic targets for reducing fibrosis.

To investigate the downstream effects of reduced stiffness-mediated FMT on ECM deposition by *Acomys* fibroblasts, *Acomys* and *Mus* CDMs were cultured over 2–3 weeks on increasing substrate stiffnesses, and compositions were quantified. This proteomics-based approach also allowed for a direct comparison between species. It is not recommended to perform cross-species comparisons during RNA sequencing due to expected differences in gene lengths and base levels

between species. However, cross-species comparisons are more common, as homology assessments are more reliable using protein sequences than DNA sequences.⁵⁷ In addition, our study focuses on secreted and ECM proteins, which are highly conserved across species.⁵⁸

Label-free quantitative proteomics of *Acomys* and *Mus* CDMs revealed that *Acomys* fibroblasts secrete significantly higher COL12A1, CEMIP, and periostin than *Mus* fibroblasts on soft PDMS, stiff PDMS, and glass. COL12A1 is a potential modulator of collagen I fibril organization,^{59,60} and overexpression of COL12A1 is associated with thinner collagen fibrils, reduced fibrosis, and increased inflammation due to increased adhesion of M1 macrophages.⁶⁰ In addition, knockout of COL12A1 in mice resulted in increased myofibroblast cell density and TGF- β signaling due to insufficient sequestration of TGF- β .⁶⁰ These results suggest that increased COL12A1 deposition by *Acomys* fibroblasts may play a role in reduced TGF- β signaling as well as promote a higher M1/M2 macrophage ratio. Interestingly, CEMIP and periostin – also upregulated in *Acomys* CDMs compared to *Mus* – have largely been associated with fibrosis. CEMIP is expressed in many different tissues and is known to degrade the ECM protein hyaluronic acid, promoting cell migra-

tion.^{61,62} Although CEMIP is most often studied in the context of cancer, CEMIP levels are higher in patients with pulmonary fibrosis, and CEMIP decreases during attenuation of fibrosis.⁶³ Periostin is most notably associated with pathogenesis of pulmonary fibrosis.⁶⁴ However, periostin is also necessary for cutaneous wound healing,^{65,66} and periostin expression is upregulated in regenerating *Acomys* wounds *in vivo* (preprint).⁶⁷ In comparison, *Mus* CDMs had higher presence of fibrillin-1, versican, COL3A1, perlecan, and fibulin-2. Each of these proteins are typically upregulated in fibrotic matrix compositions.^{68,69} For example, fibrillin-1 is elevated in rodent models of renal and cardiac fibrosis.^{70,71}

Label-free quantitative proteomics also identified unique protein compositions in *Acomys* and *Mus* CDMs. Many of the proteins unique to *Mus* CDMs are known to be associated with fibrosis. For example, ADAMTS1 was the most highly expressed protein unique to *Mus* CDMs. The ADAMTS family is a group of extracellular proteases that play a role in both ECM degradation and stabilization.^{72–74} ADAMTS1 is upregulated in hepatic,⁷⁵ cardiac,⁷⁶ and muscle⁷⁷ fibrosis, and fibrosis is attenuated upon inhibition of ADAMTS1 in mouse models. The contribution of ADAMTS1 to fibrosis progression is thought to be due in part to its direct activation of TGF- β by interacting with the latency associated peptide (LAP).^{75,76} ADAMTSL4, another member of the ADAMTS family, was also present in *Mus* CDMs but absent in *Acomys* CDMs. ADAMTSL4 associates with fibrillin and enhances fibrillin-1 deposition by cultured fibroblasts.⁷⁸ The presence of ADAMTSL4 in *Mus* CDMs may have contributed to higher fibrillin-1 production in *Mus* CDMs compared to *Acomys* CDMs observed in this study.

The matricellular proteins tenascin-X and matrilin-2 were also unique to *Mus* CDMs. Tenascin-X contributes to collagen fibrillogenesis and matrix stiffening.^{79,80} In addition, tenascin-X activates TGF- β through an integrin $\alpha 11\beta 1$ -dependent interaction with LAP, increasing TGF- β bioavailability.^{80–82} Knockout of tenascin-X reduced infiltration of myeloid cells and collagen I deposition in a mouse model of liver fibrosis.⁸³ Unlike tenascin-X, matrilin-2 knockdown has not been demonstrated to influence fibrosis progression in mouse models, although it is upregulated in fibrosis.^{84,85}

Other secreted proteins present in *Mus* CDMs but absent in *Acomys* CDMs were LGALS3BP, asporin, and ANGPTL2. LGALS3BP binds to galectins, fibronectin, and collagen, and its production is higher in patients with liver fibrosis,^{86–88} but assessments of the mechanistic role of LGALS3BP in fibrosis have not been extensively studied. Asporin is one of the most upregulated genes in fibroblasts from IPF patients and is predominantly expressed by myofibroblasts.⁸⁹ In asporin-deficient fibroblasts, expression of the TGF- β type I receptor complex was reduced and TGF- β mediated FMT was inhibited.⁸⁹ However, asporin knockdown at the organism level resulted in cardiac fibrosis following injury through negative regulation of TGF- β signaling,⁹⁰ suggesting that the role of asporin in fibrosis may be cell- or tissue-dependent and warrants further study. Increased expression of ANGPTL2 is associated with kidney,⁹¹ atrial,⁹² and liver⁹³ fibrosis. ANGPTL2 upregulates α SMA and TGF- β in cultured fibroblasts,⁹² and knockout of ANGPTL2 in a mouse model of renal fibrosis decreased TGF- β 1 amplification, macrophage infiltration, pro-inflammatory cytokines, and fibrosis.⁹¹

SPARC, galectin-1, and annexin A5 were present only in *Mus* CDMs cultured on stiff substrates, suggesting they are

upregulated by stiffness-mediated activation in *Mus* fibroblasts. SPARC is a matricellular protein that participates in collagen cross-linking, and serum SPARC concentrations increase in patients with fibrotic renal injury and systemic sclerosis.⁷⁶ Chang et al. found that SPARC expression was 8-fold greater in IPF myofibroblasts compared to fibroblasts from healthy tissue and supported myofibroblast persistence by preventing apoptosis.⁹⁴ In addition, SPARC-null mice exhibited a significant decrease in fibrotic capsule formation around a subcutaneous PDMS implant.⁹⁵ SPARC-induced collagen deposition during fibrosis is mediated by ADAMTS1,⁷⁶ which was present in *Mus* CDMs cultured on all stiffnesses in this study but absent in *Acomys* CDMs. Galectin-1 is overexpressed within keloid scar fibroblasts⁹⁶ and hypertrophic scars⁹⁷ and is known to support myofibroblast differentiation through reactive oxygen species (ROS) generation.^{98,99} Galectin-1 knockout mice have fewer α SMA positive myofibroblasts, reduced collagen deposition, and reduced macrophage infiltration in wounds.^{98,100}

While all proteins unique to *Mus* CDMs had an association with fibrosis, proteins unique to *Acomys* CDMs were less associated with fibrosis and some have been demonstrated to attenuate fibrosis. The most highly expressed ECM-related protein in *Acomys* CDMs was VWA1. VWA1 interacts with collagen VI and perlecan and is largely restricted to cartilage.^{101,102} Associations between VWA1 and fibrosis have not been reported outside of mutations in VWA1, which have been connected to neuromyopathy.^{101,103} However, VWA1 regulates FGF signaling,¹⁰⁴ which plays a role in FMT and ECM protein expression.¹⁰⁵ FGF1 and FGF2 reduce TGF- β -mediated FMT and collagen synthesis *in vitro*.^{106,107} *In vivo*, overexpression of FGF1 reduced TGF- β 1-induced pulmonary fibrosis in rats¹⁰⁶ and overexpression of FGF2 protected mice from bleomycin-induced pulmonary fibrosis.¹⁰⁸ This potential link between secretion of VWA1 by *Acomys* fibroblasts and FGF-mediated inhibition of TGF- β signaling and fibrosis warrants further investigation.

In addition to VWA1, clusterin, THSD4, SMPDL3B, ITIH5, and LOXL4 were present in *Acomys* CDMs cultured on all stiffnesses but absent in *Mus* CDMs. Clusterin is a glycoprotein expressed in most human tissues that plays a role in many biological processes.¹⁰⁹ Clusterin presence was found to be reduced in the fibrotic lung compared to healthy lungs where it was localized to fibroblasts, and *in vitro*, fibroblasts from fibrotic lungs had reduced clusterin expression.¹¹⁰ Knockout of clusterin in a mouse model of hepatic fibrosis resulted in increased fibrosis while infection with a clusterin-containing adenovirus attenuated fibrosis.¹¹¹ Overexpression of clusterin also attenuated fibrosis following a renal injury by reducing macrophage infiltration, presence of α SMA positive myofibroblasts, and collagen I and fibronectin deposition.¹¹² The ADAMTSL6 protein binds to and promotes assembly of fibrillin-1 but is also thought to be a negative regulator of TGF- β signaling.^{113,114} Local administration of ADAMTSL6 β reduced TGF- β signaling and improved wound healing in a mouse model of Marfan Syndrome.¹¹⁵ ITIH5 is a tumor suppressor that is known to bind to hyaluronic acid,¹¹⁶ but does not have a clear role in fibrosis. SMPDL3B is indirectly associated with fibrosis through stimulation of S1P levels which is linked to several fibrotic diseases,^{117,118} but there have been no assessments of the direct impact of SMPDL3B on promotion or attenuation of fibrosis. LOXL4 has associations with fibrosis due to its role in collagen cross-linking.¹¹⁹ However, inhibition

or treatment with LOXL4 does not directly impact fibroblast activation,¹¹⁹ suggesting that the role of LOXL4 in fibrosis promotion α SMA production *in vivo* is due to collagen-cross-linking which increases tissue stiffness.

Upregulation of unique proteins by *Acomys* fibroblasts cultured on stiff substrates was an unexpected outcome of this experiment due to our RNA sequencing results which demonstrated no differential gene expression in *Acomys* with response to stiffness. It is possible that prolonged culture time and production of a tissue-like environment prompted more notable changes in *Acomys* fibroblasts compared to our shorter RNA sequencing experiment. LTBP2, PCPE-1, peroxidasin, serpine-1, and TGF- β 2 were found only in *Acomys* CDMs cultured on stiff PDMS or glass substrates. LTBP2 is associated with fibrillin-1 and has roles in regulating elastic fiber assembly.^{120,121} LTBP2 cannot react with any isoforms of TGF- β and competes with LTBP1 for binding to fibrillin-1, preventing LTBP1 from activating TGF- β through release from LAP.^{122,123} This suggests that increased presence of LTBP2 may prevent TGF- β -mediated fibrotic signaling. PCPE-1, peroxidasin, serpine-1, and TGF- β 2 have more direct associations with fibrosis. PCPE-1 has roles in collagen maturation and is upregulated in several types of fibrosis.^{124–126} Peroxidasin is a peroxidase that organizes into a fibril-like network in the extracellular space.¹²⁷ It is increased in kidney and cardiac fibrosis, but the field's understanding of the roles of peroxidasin in fibrosis is still developing.^{127,128} Serpine-1 contributes to fibrosis through inhibition of matrix degradation and has been implicated in fibrosis of multiple organs, but regulation of serpine-1 during fibrogenesis can vary based on organ and cell type.¹²⁹ For example, Serpine-1 is not essential for the development of bleomycin-induced skin fibrosis, and serpine-1 deficiency does not have a protective effect.¹³⁰ TGF- β 2 expression is increased in lung and liver fibrotic tissues compared to healthy tissues and contributes to fibrosis, although its roles in fibrosis are less characterized than TGF- β 1.¹³¹ TGF- β 2 also suppresses macrophage inflammatory responses and cytokine production in response to LPS.¹³² Production of TGF- β 2 by *Acomys* fibroblasts may have contributed to the reduction in differential gene expression observed in mBMDMs cultured on *Acomys* CDMs in response to M1 stimulation.

Overall, *Acomys* and *Mus* CDMs were found to have distinct compositions, with *Mus* CDMs containing more fibrosis-associated molecules and *Acomys* CDMs containing some proteins that are known to negatively regulate fibrosis. Differing ECM compositions deposited by *Acomys* fibroblasts may play a role in the regulation of immune cells in the fibrotic environment. To assess the impact of *Acomys* and *Mus* ECM compositions on macrophage phenotype, mBMDMs were seeded on each and gene expression was assessed. Although the effect of CDM species on differential gene expression was modest, mBMDMs cultured on *Acomys* CDMs had a significant reduction in MMP9 mRNA production and enzyme activity compared to mBMDMs cultured on *Mus* CDMs. Although MMPs are typically associated with ECM degradation, MMP9 is considered a pro-fibrotic molecule, and is largely produced by M2-type macrophages.^{133–135} In a mouse model of renal fibrosis, MMP9 was upregulated in late stages of injury and was largely produced by macrophages and myofibroblasts.¹³⁶ Antibody-based inhibition of MMP9 in this model caused a reduction in macrophage infiltration and reduced staining of α SMA and collagen.¹³⁶ Similar results

occurred in MMP9 knockout mouse models of cardiac and renal fibrosis.^{137,138} Macrophages deficient in MMP9 were also found to reduce myofibroblast differentiation compared to controls *in vitro*.^{136,139} Promotion of myofibroblast differentiation by macrophage-produced MMP9 is due to the ability for MMP9 to activate latent TGF- β .^{122,139–141} Our findings suggest that the unique composition of ECM deposited by *Acomys* fibroblasts negatively regulates MMP9 production in macrophages and favors an antifibrotic macrophage phenotype which may inhibit TGF- β signaling and downstream myofibroblast activation.

Comparisons between treatment groups were limited due to sample loss. However, mBMDMs cultured on *Acomys* and *Mus* CDMs demonstrated differences in response to M1 and M2 stimulation. Specifically, mBMDMs cultured on *Acomys* CDMs had fewer differentially expressed genes following both M1 and M2 stimulation compared to mBMDMs cultured on *Mus* CDMs. In addition, the top genes upregulated in response to M1 stimulation differed between the species. These results suggest that differences in ECM composition between species differentially regulate mBMDM phenotype in response to pro-inflammatory and pro-fibrotic stimuli, but additional investigation is needed to draw any conclusions.

Protein-level quantification of Arg-1 and iNOS expression by mBMDMs seeded on *Acomys* and *Mus* CDMs revealed no changes in Arg-1 across species or treatment groups. However, iNOS expression was significantly affected by both species and stimulation treatment. In addition, when treated with IL-4, mBMDMs cultured on *Acomys* CDMs retained higher iNOS expression than those cultured on *Mus* CDMs, suggesting a resistance to IL-4 treatment.

Overall, we confirmed that *Acomys* fibroblasts were not responsive to substrate stiffness in short-term culture. In addition, the ECM produced by *Acomys* and *Mus* fibroblasts had different compositions, with *Mus* CDMs consisting of proteins that are highly associated with fibrosis. These different ECM compositions had modest effects on mBMDM phenotype which warrant further investigation.

■ ASSOCIATED CONTENT

Data Availability Statement

Scaffold DDA outputs (fold-change comparisons and identification of unique protein compositions) for all CDM group comparisons can be found at <https://data.mendeley.com/preview/jpfvfv7x9n?a=54a2d0b7-e8db-449b-9eb4-fb7ff5f5d1f2>, an open-source online data repository hosted by Mendeley Data.

■ AUTHOR INFORMATION

Corresponding Author

Erika Moore — J. Crayton Pruitt Family Department of Biomedical Engineering, University of Florida, Gainesville, Florida 32611, United States; Fischell Department of Bioengineering, University of Maryland, College Park, Maryland 20742, United States; orcid.org/0000-0003-2192-6147; Email: emt@umd.edu

Authors

Michele N. Dill — J. Crayton Pruitt Family Department of Biomedical Engineering, University of Florida, Gainesville, Florida 32611, United States; orcid.org/0000-0003-0112-3929

Zoe Turner — Fischell Department of Bioengineering,
University of Maryland, College Park, Maryland 20742,
United States

Paulina W. Kapuscinska — J. Crayton Pruitt Family
Department of Biomedical Engineering, University of Florida,
Gainesville, Florida 32611, United States

Katie Heiden — Department of Chemistry, University of
Florida, Gainesville, Florida 32611, United States

Kari B. Basso — Department of Chemistry, University of
Florida, Gainesville, Florida 32611, United States;
orcid.org/0000-0003-1268-6655

Chelsey S. Simmons — Department of Mechanical and
Aerospace Engineering, University of Florida, Gainesville,
Florida 32611, United States; Present Address: Bain and
Company

Complete contact information is available at:

<https://pubs.acs.org/10.1021/acsbiomaterials.5c00810>

Author Contributions

M.N.D.: conceptualization, formal analysis, investigation, methodology, visualization, writing—original draft preparation. Z.T.: formal analysis, investigation, writing - review and editing. P.W.K.: investigation, writing—review and editing. K.H.: formal analysis, investigation, writing—review and editing. K.B.B.: formal analysis, funding acquisition, visualization, writing—review and editing. C.S.S.: conceptualization, funding acquisition, supervision, writing—review and editing. E.M.: project administration, supervision, writing—review and editing.

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

This work was supported by the National Science Foundation (NSF 1636007) and the National Institutes of Health (NIH R35GM128831, NIH S10 OD021758-01A1, and NIH S10 OD030250-01A1). This material is based upon work supported by the National Science Foundation Graduate Research Fellowship Program under Grant No. DGE—1842473. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the National Science Foundation. The authors thank the University of Florida ICBR Gene Expression and Genotyping Core Facility, RRID:SCR_019145 for performing RNA Sequencing and the University of Florida ICBR Bioinformatics Core Facility, RRID:SCR_019120 for assistance with RNA Sequencing data analysis. Figures ¹A, ²A, and ⁵A, and the graphical abstract were created in <https://BioRender.com>.

ABBREVIATIONS

CDMs	cell-derived matrices
ECM	extracellular matrix
FMT	fibroblast-to-myofibroblast transition
mBMDMs	mouse bone marrow-derived macrophages
PDMS	polydimethylsiloxane
TCP	tissue culture plastic

REFERENCES

(1) Wynn, T. A. Fibrotic Disease and the T(H)1/T(H)2 Paradigm. *Nat. Rev. Immunol.* **2004**, *4* (8), 583–594.

- (2) Henderson, N. C.; Rieder, F.; Wynn, T. A. FIBROSIS: FROM MECHANISMS TO MEDICINES. *Nature* **2020**, 587 (7835), 555.
- (3) Seifert, A. W.; Kiama, S. G.; Seifert, M. G.; Goheen, J. R.; Palmer, T. M.; Maden, M. Skin Shedding and Tissue Regeneration in African Spiny Mice (*Acomys*). *Nature* **2012**, 489 (7417), 561–565.
- (4) Brant, J. O.; Lopez, M. C.; Baker, H. V.; Barbazuk, W. B.; Maden, M. A Comparative Analysis of Gene Expression Profiles during Skin Regeneration in Mus and *Acomys*. *PLoS One* **2015**, *10* (11), No. e0142931.
- (5) Maden, M.; Brant, J. O. Insights into the Regeneration of Skin from *Acomys*, the Spiny Mouse. *Exp. Dermatol.* **2019**, *28* (4), 436–441.
- (6) Yoon, J. H.; Cho, K.; Garrett, T. J.; Finch, P.; Maden, M. Comparative Proteomic Analysis in Scar-Free Skin Regeneration in *Acomys Cahirinus* and Scarring Mus *Musculus*. *Sci. Rep.* **2020**, *10* (1), 1–16.
- (7) Brant, J. O.; Yoon, J. H.; Polvadore, T.; Barbazuk, W. B.; Maden, M. Cellular Events during Scar-Free Skin Regeneration in the Spiny Mouse, *Acomys*. *Wound Repair Regen.* **2016**, *24* (1), 75–88.
- (8) Maden, M. Optimal Skin Regeneration after Full Thickness Thermal Burn Injury in the Spiny Mouse, *Acomys Cahirinus*. *Burns* **2018**, *44* (6), 1509–1520.
- (9) Matias Santos, D.; Martins Rita, A.; Casanellas, I.; Brito Ova, A.; Maria Ará Ujo, I.; Power, D.; Tiscornia, G. Ear Wound Regeneration in the African Spiny Mouse *Acomys Cahirinus*. *Regeneration* **2016**, *3* (1), 52–61.
- (10) Gawriluk, T. R.; Simkin, J.; Thompson, K. L.; Biswas, S. K.; Clare-Salzler, Z.; Kimani, J. M.; Kiama, S. G.; Smith, J. J.; Ezenwa, V. O.; Seifert, A. W. Comparative Analysis of Ear-Hole Closure Identifies Epimorphic Regeneration as a Discrete Trait in Mammals. *Nat. Commun.* **2016**, *7* (1), 1–16.
- (11) Simkin, J.; Gawriluk, T. R.; Gensel, J. C.; Seifert, A. W. Macrophages Are Necessary for Epimorphic Regeneration in African Spiny Mice. *Elife* **2017**, *6*, No. e24623.
- (12) Maden, M.; Brant, J. O.; Rubiano, A.; Sandoval, A. G. W.; Simmons, C.; Mitchell, R.; Collin-Hooper, H.; Jacobson, J.; Omairi, S.; Patel, K. Perfect Chronic Skeletal Muscle Regeneration in Adult Spiny Mice, *Acomys Cahirinus*. *Sci. Rep.* **2018**, *8* (1), 1–14.
- (13) Okamura, D. M.; Brewer, C. M.; Wakenight, P.; Bahrami, N.; Bernardi, K.; Tran, A.; Olson, J.; Shi, X.; Yeh, S.-Y.; Piliponsky, A.; Collins, S. J. Spiny Mice Activate Unique Transcriptional Programs after Severe Kidney Injury Regenerating Organ Function without Fibrosis. *iScience* **2021**, *24* (11), 103269.
- (14) Peng, H.; Shindo, K.; Donahue, R. R.; Gao, E.; Ahern, B. M.; Levitan, B. M.; Tripathi, H.; Powell, D.; Noor, A.; Elmore, G. A.; Satin, J.; Seifert, A. W.; Abdel-Latif, A. Adult Spiny Mice (*Acomys*) Exhibit Endogenous Cardiac Recovery in Response to Myocardial Infarction. *Npj Regen. Med.* **2021**, *6* (1), 1–15.
- (15) Koopmans, T.; van Beijnum, H.; Roovers, E. F.; Tomasso, A.; Malhotra, D.; Boeter, J.; Psathaki, O. E.; Versteeg, D.; van Rooij, E.; Bartscherer, K. Ischemic Tolerance and Cardiac Repair in the Spiny Mouse (*Acomys*). *Npj Regen. Med.* **2021**, *6* (1), 1–16.
- (16) Streeter, K. A.; Sunshine, M. D.; Brant, J. O.; Sandoval, A. G. W.; Maden, M.; Fuller, D. D. Molecular and Histologic Outcomes Following Spinal Cord Injury in Spiny Mice, *Acomys Cahirinus*. *J. Comp. Neurol.* **2020**, *528* (9), 1535–1547.
- (17) Nogueira-Rodrigues, J.; Leite, S. C.; Pinto-Costa, R.; Sousa, S. C.; Luz, L. L.; Sintra, M. A.; Oliveira, R.; Monteiro, A. C.; Pinheiro, G. G.; Vitorino, M.; Silva, J. A.; Simão, S.; Fernandes, V. E.; Provazník, J.; Benes, V.; Cruz, C. D.; Safronov, B. V.; Magalhães, A.; Reis, C. A.; Vieira, J.; Vieira, C. P.; Tiscornia, G.; Araújo, I. M.; Sousa, M. M. Rewired Glycosylation Activity Promotes Scarless Regeneration and Functional Recovery in Spiny Mice after Complete Spinal Cord Transection. *Dev. Cell* **2022**, *57*, 440.
- (18) D'Urso, M.; Kurniawan, N. A. Mechanical and Physical Regulation of Fibroblast-Myofibroblast Transition: From Cellular Mechanoreponse to Tissue Pathology. *Front. Bioeng. Biotechnol.* **2020**, *8*, 609653.

- (19) Zhang, H.; Zhou, Y.; Wen, D.; Wang, J. Noncoding RNAs: Master Regulator of Fibroblast to Myofibroblast Transition in Fibrosis. *Int. J. Mol. Sci.* **2023**, *24* (2), 1801.
- (20) Kanisicak, O.; Khalil, H.; Ivey, M. J.; Karch, J.; Maliken, B. D.; Correll, R. N.; Brody, M. J.; Lin, S. C. J.; Aronow, B. J.; Tallquist, M. D.; Molkentin, J. D. Genetic Lineage Tracing Defines Myofibroblast Origin and Function in the Injured Heart. *Nat. Commun.* **2016**, *7* (1), 1–14.
- (21) Geng, Y.; Liu, X.; Liang, J.; Habel, D. M.; Vrishika, K.; Coelho, A. L.; Deng, N.; Xie, T.; Wang, Y.; Liu, N.; et al. PD-L1 on Invasive Fibroblasts Drives Fibrosis in a Humanized Model of Idiopathic Pulmonary Fibrosis. *JCI Insight* **2019**, *4* (6), No. e125326.
- (22) Lin, H.; Wu, C.; Zhu, F.; Zhang, G.; Xie, Y.; Cui, Y.; Dong, G.; Zhang, X. Anti-Fibrotic Effect of Igaratimod on Pulmonary Fibrosis by Inhibiting the Fibroblast-to-Myofibroblast Transition. *Adv. Med. Sci.* **2020**, *65* (2), 338–347.
- (23) Liu, Y.; Chen, S.; Yu, L.; Deng, Y.; Li, D.; Yu, X.; Chen, D.; Lu, Y.; Liu, S.; Chen, R. Pemafibrate Attenuates Pulmonary Fibrosis by Inhibiting Myofibroblast Differentiation. *Int. Immunopharmacol.* **2022**, *108*, 108728.
- (24) Wynn, T. A. Cellular and Molecular Mechanisms of Fibrosis. *J. Pathol.* **2008**, *214* (2), 199–210.
- (25) Walraven, M.; Hinz, B. Therapeutic Approaches to Control Tissue Repair and Fibrosis: Extracellular Matrix as a Game Changer. *Matrix Biol.* **2018**, *71–72*, 205–224.
- (26) Tomasek, J. J.; Gabbiani, G.; Hinz, B.; Chaponnier, C.; Brown, R. A. Myofibroblasts and Mechano-Regulation of Connective Tissue Remodelling. *J. Comp. Neurol.* **2002**, *3* (5), 349–363.
- (27) Wynn, T. A. Common and Unique Mechanisms Regulate Fibrosis in Various Fibroproliferative Diseases. *J. Clin. Invest.* **2007**, *117*, 524–529.
- (28) Wilson, M. S.; Elnekave, E.; Mentink-Kane, M. M.; Hodges, M. G.; Pesce, J. T.; Ramalingam, T. R.; Thompson, R. W.; Kamanaka, M.; Flavell, R. A.; Keane-Myers, A.; Cheever, A. W.; Wynn, T. A. IL-13R α 2 and IL-10 Coordinately Suppress Airway Inflammation, Airway-Hyperreactivity, and Fibrosis in Mice. *J. Clin. Invest.* **2007**, *117* (10), 2941–2951.
- (29) Liu, F.; Mih, J. D.; Shea, B. S.; Kho, A. T.; Sharif, A. S.; Tager, A. M.; Tschumperlin, D. J. Feedback Amplification of Fibrosis through Matrix Stiffening and COX-2 Suppression. *J. Cell Biol.* **2010**, *190* (4), 693–706.
- (30) Noskovicova, N.; Schuster, R.; van Putten, S.; Ezzo, M.; Koehler, A.; Boo, S.; Coelho, N. M.; Griggs, D.; Ruminski, P.; McCulloch, C. A.; Hinz, B. Suppression of the Fibrotic Encapsulation of Silicone Implants by Inhibiting the Mechanical Activation of Pro-Fibrotic TGF- β . *Nat. Biomed. Engineering* **2021**, *5* (12), 1437–1456.
- (31) Hinz, B. The Myofibroblast: Paradigm for a Mechanically Active Cell. *J. Biomech* **2010**, *43* (1), 146–155.
- (32) Janmey, P. A.; Fletcher, D. A.; Reinhart-King, C. A. Stiffness Sensing by Cells. *Physiol. Rev.* **2020**, *100* (2), 695.
- (33) Pakshir, P.; Hinz, B. The Big Five in Fibrosis: Macrophages, Myofibroblasts, Matrix, Mechanics, and Miscommunication. *Matrix Biol.* **2018**, *68–69*, 81–93.
- (34) Lagares, D.; Busnadiego, O.; García-Fernández, R. A.; Kapoor, M.; Liu, S.; Carter, D. E.; Abraham, D.; Shi-Wen, X.; Carreira, P.; Fontaine, B. A.; Shea, B. S.; Tager, A. M.; Leask, A.; Lamas, S.; Rodríguez-Pascual, F. Inhibition of Focal Adhesion Kinase Prevents Experimental Lung Fibrosis and Myofibroblast Formation. *Arthritis Rheumatol* **2012**, *64* (5), 1653.
- (35) Liang, M.; Yu, M.; Xia, R.; Song, K.; Wang, J.; Luo, J.; Chen, G.; Cheng, J. Yap/Taz Deletion in Gli⁺ Cell-Derived Myofibroblasts Attenuates Fibrosis. *J. Am. Soc. Nephrol.* **2017**, *28* (11), 3278–3290.
- (36) Brewer, C. M.; Nelson, B. R.; Wakenight, P.; Collins, S. J.; Okamura, D. M.; Dong, X. R.; Mahoney, W. M.; McKenna, A.; Shendure, J.; Timms, A.; Millen, K. J.; Majesky, M. W. Adaptations in Hippo-Yap Signaling and Myofibroblast Fate Underlie Scar-Free Ear Appendage Wound Healing in Spiny Mice. *Dev. Cell.* **2021**, *56* (19), 2722.
- (37) Gaire, J.; Dill, M.; Supper, V.; Simmons, C. S.; Crayton Pruitt, J.; Box, P. O. Attenuated Foreign Body Response to Subcutaneous Implant in Regenerative Spiny Mice (*Acomys*). *bioRxiv* **2022**, *8* (12), 503776.
- (38) Stewart, D. C.; Serrano, P. N.; Rubiano, A.; Yokosawa, R.; Sandler, J.; Mukhtar, M.; Brant, J. O.; Maden, M.; Simmons, C. S. Unique Behavior of Dermal Cells from Regenerative Mammal, the African Spiny Mouse, in Response to Substrate Stiffness. *J. Biomech.* **2018**, *81*, 149–154.
- (39) Wynn, T. A.; Chawla, A.; Pollard, J. W. Macrophage Biology in Development, Homeostasis and Disease. *Nature* **2013**, *496* (7446), 445–455.
- (40) Kim, S. Y.; Nair, M. G. Macrophages in Wound Healing: Activation and Plasticity. *Immunol. Cell Biol.* **2019**, *97* (3), 258.
- (41) Huleihel, L.; Dziki, J. L.; Bartolacci, J. G.; Rausch, T.; Scarratt, M. E.; Cramer, M. C.; Vorobyov, T.; LoPresti, S. T.; Swineheart, I. T.; White, L. J.; Brown, B. N.; Badylak, S. F. Macrophage Phenotype in Response to ECM Bioscaffolds. *Semin. Immunol.* **2017**, *29*, 2.
- (42) Dziki, J. L.; Wang, D. S.; Pineda, C.; Sicari, B. M.; Rausch, T.; Badylak, S. F. Solubilized Extracellular Matrix Bioscaffolds Derived from Diverse Source Tissues Differentially Influence Macrophage Phenotype. *J. Biomed Mater. Res. A* **2017**, *105* (1), 138–147.
- (43) Haschak, M.; LoPresti, S.; Stahl, E.; Dash, S.; Popovich, B.; Brown, B. N. Macrophage Phenotype and Function Are Dependent upon the Composition and Biomechanics of the Local Cardiac Tissue Microenvironment. *Aging* **2021**, *13* (13), 16938.
- (44) Puttock, E. H.; Tyler, E. J.; Manni, M.; Maniati, E.; Butterworth, C.; Burger Ramos, M.; Peerani, E.; Hirani, P.; Gauthier, V.; Liu, Y.; Maniscalco, G.; Rajeev, V.; Cutillas, P.; Trevisan, C.; Pozzobon, M.; Lockley, M.; Rastrick, J.; Läubli, H.; White, A.; Pearce, O. M. T. Extracellular Matrix Educates an Immunoregulatory Tumor Macrophage Phenotype Found in Ovarian Cancer Metastasis. *Nat. Commun.* **2023**, *14* (1), 1–16.
- (45) Simkin, J.; Aloysius, A.; Adam, M.; Safaee, F.; Donahue, R. R.; Biswas, S.; Lakhani, Z.; Gensel, J. C.; Thybert, D.; Potter, S.; Seifert, A. W. Tissue-Resident Macrophages Specifically Express Lactotransferrin and Vegfc during Ear Pinna Regeneration in Spiny Mice. *Dev. Cell* **2024**, *59* (4), 496–516.e6.
- (46) Ong, C.; Wong, C.; Roberts, C. R.; Teh, H.-S.; Jirik, F. R. Anti-IL-4 Treatment Prevents Dermal Collagen Deposition in the Tight-skin Mouse Model of Scleroderma. *Eur. J. Immunol.* **1998**, *28* (9), 2619–2629.
- (47) Weng, S. Y.; Wang, X.; Vijayan, S.; Tang, Y.; Kim, Y. O.; Padberg, K.; Regen, T.; Molokanova, O.; Chen, T.; Bopp, T.; Schild, H.; Brombacher, F.; Crosby, J. R.; McCaleb, M. L.; Waisman, A.; Bockamp, E.; Schuppan, D. IL-4 Receptor Alpha Signaling through Macrophages Differentially Regulates Liver Fibrosis Progression and Reversal. *EBioMedicine* **2018**, *29*, 92.
- (48) Nguyen, J. K.; Austin, E.; Huang, A.; Mamalis, A.; Jagdeo, J. The IL-4/IL-13 Axis in Skin Fibrosis and Scarring: Mechanistic Concepts and Therapeutic Targets. *Arch. Dermatol. Res.* **2020**, *312* (2), 81.
- (49) Fitzpatrick, L. E.; McDevitt, T. C. Cell-Derived Matrices for Tissue Engineering and Regenerative Medicine Applications. *J. Biomater. Sci.* **2015**, *3* (1), 12.
- (50) Palchesko, R. N.; Zhang, L.; Sun, Y.; Feinberg, A. W. Development of Polydimethylsiloxane Substrates with Tunable Elastic Modulus to Study Cell Mechanobiology in Muscle and Nerve. *PLoS One* **2012**, *7* (12), No. e51499.
- (51) Sikulskyi, S.; Ren, Z.; Mekonnen, D. T.; Holyoak, A.; Srinivasaraghavan Govindarajan, R.; Kim, D. Additively Manufactured Unimorph Dielectric Elastomer Actuators: Design, Materials, and Fabrication. *Front. Robot. AI* **2022**, *9*, 1034914.
- (52) Nguyen, E. D.; Fard, V. N.; Kim, B. Y.; Collins, S.; Galey, M.; Nelson, B. R.; Wakenight, P.; Gable, S. M.; McKenna, A.; Bammler, T. K.; et al. Genome Report: Chromosome-Scale Genome Assembly of the African Spiny Mouse (*Acomys Cahirinus*). *Eur. J. Immunol.* **2023**, *13* (10), 535372.

- (53) Harris, G. M.; Raitman, I.; Schwarzbauer, J. E. Cell-Derived Decellularized Extracellular Matrices. *Methods Cell Biol.* **2018**, *143*, 97.
- (54) Zeiger, A. S.; Loe, F. C.; Li, R.; Raghunath, M.; van Vliet, K. J. Macromolecular Crowding Directs Extracellular Matrix Organization and Mesenchymal Stem Cell Behavior. *PLoS One* **2012**, *7* (5), 37904.
- (55) Zhou, Y.; Huang, X.; Hecker, L.; Kurundkar, D.; Kurundkar, A.; Liu, H.; Jin, T. H.; Desai, L.; Bernard, K.; Thannickal, V. J. Inhibition of Mechanosensitive Signaling in Myofibroblasts Ameliorates Experimental Pulmonary Fibrosis. *J. Clin. Invest.* **2013**, *123* (3), 1096–1108.
- (56) Lagares, D.; Santos, A.; Grasberger, P. E.; Liu, F.; Probst, C. K.; Rahimi, R. A.; Sakai, N.; Kuehl, T.; Ryan, J.; Bhola, P.; et al. Targeted Apoptosis of Myofibroblasts with the BH3Mimetic ABT-263 Reverses Established Fibrosis. *Sci. Transl. Med.* **2017**, *9* (420), No. eaal3765.
- (57) Pearson, W. R. An Introduction to Sequence Similarity (“Homology”) Searching. *Curr. Protoc. Bioinformatics.* **2013**, *3* (SUPPL.42), 1–3.
- (58) Özbek, S.; Balasubramanian, P. G.; Chiquet-Ehrismann, R.; Tucker, R. P.; Adams, J. C. The Evolution of Extracellular Matrix. *Mol. Biol. Cell* **2010**, *21* (24), 4300.
- (59) Izu, Y.; Birk, D. E. Collagen XII Mediated Cellular and Extracellular Mechanisms in Development, Regeneration, and Disease. *Front. Cell Dev. Biol.* **2023**, *11*, 1129000.
- (60) Schönborn, K.; Willenborg, S.; Schulz, J. N.; Imhof, T.; Eming, S. A.; Quondamatteo, F.; Brinckmann, J.; Niehoff, A.; Paulsson, M.; Koch, M.; Eckes, B.; Krieg, T. Role of Collagen XII in Skin Homeostasis and Repair. *Matrix Biol* **2020**, *94*, 57–76.
- (61) Domanegg, K.; Sleeman, J. P.; Schmaus, A. CEMIP, a Promising Biomarker That Promotes the Progression and Metastasis of Colorectal and Other Types of Cancer. *Cancers* **2022**, *14* (20), 5093.
- (62) Spataro, S.; Guerra, C.; Cavalli, A.; Sgrignani, J.; Sleeman, J.; Poulain, L.; Boland, A.; Scapozza, L.; Moll, S.; Prunotto, M. CEMIP (HYBID, KIAA1199): Structure, Function and Expression in Health and Disease. *Febs J* **2023**, *290* (16), 3946–3962.
- (63) Kwapiszewska, G.; Gungl, A.; Wilhelm, J.; Marsh, L. M.; Puthenparampil, H. T.; Sinn, K.; Didiasova, M.; Klepetko, W.; Kosanovic, D.; Schermuly, R. T.; et al. Transcriptome Profiling Reveals the Complexity of Pirfenidone Effects in Idiopathic Pulmonary Fibrosis. *Eur. Respir. J* **2018**, *52* (5), 1800564.
- (64) O'Dwyer, D. N.; Moore, B. B. The Role of Periostin in Lung Fibrosis and Airway Remodeling. *Cell. Mol. Life Sci* **2017**, *74* (23), 4305.
- (65) Nishiyama, T.; Kii, I.; Kashima, T. G.; Kikuchi, Y.; Ohazama, A.; Shimazaki, M.; Fukayama, M.; Kudo, A.; Agarwal, S. Delayed Re-Epithelialization in Periostin-Deficient Mice during Cutaneous Wound Healing. *PLoS One* **2011**, *6* (4), No. e18410.
- (66) Walker, J. T.; McLeod, K.; Kim, S.; Conway, S. J.; Hamilton, D. W. Periostin as a Multifunctional Modulator of the Wound Healing Response. *Cell Tissue Res* **2016**, *365* (3), 453–465.
- (67) McCabe, M. C.; Okamura, D. M.; Erickson, C. B.; Perry, B. W.; Brewer, C. M.; Nguyen, E. D.; Saviola, A. J.; Majesky, M. W.; Hansen, K. C. ECM-Focused Proteomic Analysis of Ear Punch Regeneration in *Acomys Cahirinus*. *bioRxiv* **2023**, *10*, 561940.
- (68) Booth, A. J.; Hadley, R.; Cornett, A. M.; Dreffs, A. A.; Matthes, S. A.; Tsui, J. L.; Weiss, K.; Horowitz, J. C.; Fiore, V. F.; Barker, T. H.; Moore, B. B.; Martinez, F. J.; Niklason, L. E.; White, E. S. Acellular Normal and Fibrotic Human Lung Matrices as a Culture System for in Vitro Investigation. *Am. J. Respir. Crit. Care Med* **2012**, *186* (9), 866–876.
- (69) Lord, M. S.; Tang, F.; Rnjak-Kovacina, J.; Smith, J. G. W.; Melrose, J.; Whitelock, J. M. The Multifaceted Roles of Perlecan in Fibrosis. *Matrix Biol* **2018**, *68–69*, 150–166.
- (70) Bouzeghrane, F.; Reinhardt, D. P.; Reudelhuber, T. L.; Thibault, G. Enhanced Expression of Fibrillin-1, a Constituent of the Myocardial Extracellular Matrix in Fibrosis. *Am. J. Physiol. Heart Circ. Physiol* **2005**, *289* (3), H982–H991.
- (71) Li, L.; Huang, J.; Liu, Y. The Extracellular Matrix Glycoprotein Fibrillin-1 in Health and Disease. *Front. Cell Dev. Biol* **2024**, *11*, 1302285.
- (72) Kuno, K.; Matsushima, K. ADAMTS-1 Protein Anchors at the Extracellular Matrix through the Thrombospondin Type I Motifs and Its Spacing Region. *J. Biol. Chem* **1998**, *273* (22), 13912–13917.
- (73) Mead, T. J.; Apte, S. S. ADAMTS Proteins in Human Disorders. *Matrix Biol* **2018**, *71–72*, 225.
- (74) Perrucci, G. L.; Rurali, E.; Pompilio, G. Cardiac Fibrosis in Regenerative Medicine: Destroy to Rebuild. *J. Thorac. Dis* **2018**, *10* (Suppl 20), S2376.
- (75) Bourd-Boittin, K.; Bonnier, D.; Leyme, A.; Mari, B.; Tuffery, P.; Samson, M.; Ezan, F.; Baffet, G.; Theret, N. Protease Profiling of Liver Fibrosis Reveals the ADAM Metalloproteinase with Thrombospondin Type 1 Motif, 1 as a Central Activator of Transforming Growth Factor Beta. *Hepatology* **2010**, *54* (6), 2173–2184.
- (76) Toba, H.; de Castro Brás, L. E.; Baicu, C. F.; Zile, M. R.; Lindsey, M. L.; Bradshaw, A. D. Increased ADAMTS1 Mediates SPARC-Dependent Collagen Deposition in the Aging Myocardium. *Am. J. Physiol. Endocrinol. Metab* **2016**, *310* (11), No. E1027–E1035.
- (77) Wang, Y.; Xiao, Y.; Zheng, Y.; Yang, L.; Wang, D. An Anti-ADAMTS1 Treatment Relieved Muscle Dysfunction and Fibrosis in Dystrophic Mice. *Life Sci* **2021**, *281*, 119756.
- (78) Gabriel, Luis A. R.; Wang, L. W.; Bader, H.; Ho, J. C.; Majors, A. K.; Hollyfield, J. G.; Traboulsi, E. I.; Apte, S. S. ADAMTS1, a Secreted Glycoprotein Widely Distributed in the Eye, Binds Fibrillin-1 Microfibrils and Accelerates Microfibril Biogenesis. *Invest. Ophthalmol. Vis. Sci* **2012**, *53* (1), 461–469.
- (79) Margaron, Y.; Bostan, L.; Exposito, J. Y.; Malbouyres, M.; Trunfio-Sfarghiu, A. M.; Berthier, Y.; Lethias, C. Tenascin-X Increases the Stiffness of Collagen Gels without Affecting Fibrillogenesis. *Biophys. Chem* **2010**, *147* (1–2), 87–91.
- (80) Imanaka-Yoshida, K.; Matsumoto, K. Multiple Roles Of Tenascins In Homeostasis And Pathophysiology Of Aorta. *Ann. Vasc. Dis.* **2018**, *11* (2), 169.
- (81) Kasprzycka, M.; Hammarström, C.; Haraldsen, G. Tenascins in Fibrotic Disorders—from Bench to Bedside. *Cell Adh. Migr* **2015**, *9* (1–2), 83.
- (82) Alcaraz, L. B.; Exposito, J. Y.; Chuvp, N.; Pommier, R. M.; Cluzel, C.; Martel, S.; Sentis, S.; Bartholin, L.; Lethias, C.; Valcourt, U. Tenascin-X Promotes Epithelial-to-Mesenchymal Transition by Activating Latent TGF- β . *J. Cell Biol* **2014**, *205* (3), 409.
- (83) Yamaguchi, S.; Kawakami, K.; Satoh, K.; Fukunaga, N.; Akama, K.; Matsumoto, K. I. Suppression of Hepatic Dysfunction in Tenascin-X-Deficient Mice Fed a High-Fat Diet. *Mol. Med. Rep* **2017**, *16* (4), 4061.
- (84) Hintermann, E.; Bayer, M.; Pfeilschifter, J. M.; Deák, F.; Kiss, I.; Paulsson, M.; Christen, U. Upregulation of Matrilin-2 Expression in Murine Hepatic Stellate Cells during Liver Injury Has No Effect on Fibrosis Formation and Resolution. *Liver Int.* **2015**, *35* (4), 1265–1273.
- (85) Szabó, E.; Korpos, É.; Batmunkh, E.; Lotz, G.; Holczbauer, Á.; Kovács, I.; Deák, F.; Kiss, I.; Schaff, Z.; Kiss, A. Expression of Matrilin-2 in Liver Cirrhosis and Hepatocellular Carcinoma. *Pathol. Oncol. Res.* **2008**, *14* (1), 15–22.
- (86) Cheung, K. J.; Libbrecht, L.; Tillemann, K.; Deforce, D.; Colle, I.; Van Vlierberghe, H. Galectin-3-Binding Protein: A Serological and Histological Assessment in Accordance with Hepatitis C-Related Liver Fibrosis. *Eur. J. Gastroenterol. Hepatol.* **2010**, *22* (9), 1066–1073.
- (87) Kuno, A.; Ikehara, Y.; Tanaka, Y.; Ito, K.; Matsuda, A.; Sekiya, S.; Hige, S.; Sakamoto, M.; Kage, M.; Mizokami, M.; Narimatsu, H. A Serum “Sweet-Doughnut” Protein Facilitates Fibrosis Evaluation and Therapy Assessment in Patients with Viral Hepatitis. *Sci. Rep* **2013**, *3*, 1065.
- (88) White, M. J. V.; Roife, D.; Gomer, R. H. Galectin-3 Binding Protein Secreted by Breast Cancer Cells Inhibits Monocyte-Derived Fibrocyte Differentiation. *J. Immunol* **2015**, *195* (4), 1858.

- (89) Huang, S.; Lai, X.; Yang, L.; Ye, F.; Huang, C.; Qiu, Y.; Lin, S.; Pu, L.; Wang, Z.; Huang, W. Asporin Promotes TGF- β -Induced Lung Myofibroblast Differentiation by Facilitating Rab11-Dependent Recycling of T β RI. *Am. J. Respir. Cell Mol. Biol.* **2022**, *66* (2), 158–170.
- (90) Huang, C.; Sharma, A.; Thakur, R.; Rai, D.; Katiki, M.; Germano, J.; Song de, Y.; Singh, S.; Sin, J.; Sengstock, D.; Andres, A. M.; Murali, R.; Mentzer, R. M.; Gottlieb, R. A.; Piplani, H. Asporin, an Extracellular Matrix Protein, Is a Beneficial Regulator of Cardiac Remodeling. *Matrix Biol.* **2022**, *110*, 40–59.
- (91) Morinaga, J.; Kadomatsu, T.; Miyata, K.; Endo, M.; Terada, K.; Tian, Z.; Sugizaki, T.; Tanigawa, H.; Zhao, J.; Zhu, S.; Sato, M.; Araki, K.; Iyama, K. I.; Tomita, K.; Mukoyama, M.; Tomita, K.; Kitamura, K.; Oike, Y. Angiopoietin-like Protein 2 Increases Renal Fibrosis by Accelerating Transforming Growth Factor- β Signaling in Chronic Kidney Disease. *Kidney Int.* **2016**, *89* (2), 327–341.
- (92) Kira, S.; Abe, I.; Ishii, Y.; Miyoshi, M.; Oniki, T.; Arakane, M.; Daa, T.; Teshima, Y.; Yufu, K.; Shimada, T.; Takahashi, N. Role of Angiopoietin-like Protein 2 in Atrial Fibrosis Induced by Human Epicardial Adipose Tissue: Analysis Using an Organo-Culture System. *Heart Rhythm* **2020**, *17* (9), 1591–1601.
- (93) Deng, Y.; Zhao, H.; Zhou, J.; Yan, L.; Wang, G.; Ma, A. L.; Xie, S. Bin; Zhang, X. Q.; Zhang, D. Z.; Xie, Q.; Zhang, G.; Shang, J.; Chen, J.; Zhao, W. F.; Zhou, Z. Q.; Liu, Y. X.; Zhang, Z. Q.; Peng, X.; Sun, X. W.; Zhang, M. X.; Lv, F. D.; Li, J.; Sun, W. L.; Bai, L.; Chen, L. M.; Liu, D.; Xi, H. L.; Zhou, Y. H.; Yu, M. Angiopoietin-like Protein as a Novel Marker for Liver Fibrosis in Chronic Hepatitis B Patients with Normal to Minimally Raised ALT. *BMC Infect. Dis.* **2017**, *17* (1), 1–9.
- (94) Chang, W.; Wei, K.; Jacobs, S. S.; Upadhyay, D.; Weill, D.; Rosen, G. D. SPARC Suppresses Apoptosis of Idiopathic Pulmonary Fibrosis Fibroblasts through Constitutive Activation of Beta-Catenin. *J. Biol. Chem.* **2010**, *285* (11), 8196–8206.
- (95) Puolakkainen, P.; Bradshaw, A. D.; Kyriakides, T. R.; Reed, M.; Brekken, R.; Wight, T.; Bornstein, P.; Ratner, B.; Sage, E. H. Compromised Production of Extracellular Matrix in Mice Lacking Secreted Protein, Acidic and Rich in Cysteine (SPARC) Leads to a Reduced Foreign Body Reaction to Implanted Biomaterials. *Am. J. Pathol.* **2003**, *162* (2), 627–635.
- (96) Arciniegas, E.; Carrillo, L.; Rojas, H.; Ramirez, R.; Martinez, I.; Rodriguez, A.; Pineda, J. Presence of Galectin-1 in the Epidermal and Dermal Thickening of Keloid Tissues. *Am. J. Res. Med. Sci.* **2017**, *1* (1), 38.
- (97) Kirkpatrick, L. D.; Shupp, J. W.; Smith, R. D.; Alkhalil, A.; Moffatt, L. T.; Carney, B. C. Galectin-1 Production Is Elevated in Hypertrophic Scar. *Wound Repair Regen.* **2021**, *29* (1), 117–128.
- (98) Lin, Y. Te; Chen, J. S.; Wu, M. H.; Hsieh, I. S.; Liang, C. H.; Hsu, C. L.; Hong, T. M.; Chen, Y. L. Galectin-1 Accelerates Wound Healing by Regulating the Neuropilin-1/Smad3/NOX4 Pathway and ROS Production in Myofibroblasts. *J. Invest. Dermatol.* **2015**, *135* (1), 258–268.
- (99) Dvořáková, B.; Szabo, P.; Lacina, L.; Gal, P.; Uhrova, J.; Zima, T.; Kaltner, H.; André, S.; Gabius, H. J.; Sykova, E.; Smetana, K. Human Galectins Induce Conversion of Dermal Fibroblasts into Myofibroblasts and Production of Extracellular Matrix: Potential Application in Tissue Engineering and Wound Repair. *Cells Tissues Organs* **2011**, *194* (6), 469–480.
- (100) Yaseen, H.; Butenko, S.; Polishuk-Zotkin, I.; Schiff-Zuck, S.; Pérez-Sáez, J. M.; Rabinovich, G. A.; Ariel, A. Galectin-1 Facilitates Macrophage Reprogramming and Resolution of Inflammation Through IFN- β . *Front. Pharmacol.* **2020**, *11*, 901.
- (101) Deschauer, M.; Hengel, H.; Rupprich, K.; Kreiß, M.; Schlotter-Weigel, B.; Grimm, M.; Admard, J.; Schneider, I.; Alhaddad, B.; Gazou, A.; Sturm, M.; Vorgerd, M.; Balousha, G.; Balousha, O.; Falna, M.; Kirschke, J. S.; Kornblum, C.; Jordan, B.; Kraya, T.; Strom, T. M.; Weis, J.; Schols, L.; Schara, U.; Zierz, S.; Riess, O.; Meitinger, T.; Haack, T. B. Bi-Allelic Truncating Mutations in VWA1 Cause Neuromyopathy. *Brain* **2021**, *144* (2), 574–583.
- (102) Allen, J. M.; Zamurs, L.; Brachvogel, B.; Schlötzer-Schrehardt, U.; Hansen, U.; Lamandé, S. R.; Rowley, L.; Fitzgerald, J.; Bateman, J. F. Mice Lacking the Extracellular Matrix Protein WARP Develop Normally but Have Compromised Peripheral Nerve Structure and Function. *J. Biol. Chem.* **2009**, *284* (18), 12020.
- (103) Pagnamenta, A. T.; Kaiyrzhanov, R.; Zou, Y.; Da'as, S. I.; Maroofian, R.; Donkervoort, S.; Dominik, N.; Lauffer, M.; Ferla, M. P.; Orioli, A.; et al. An ancestral 10-bp repeat expansion in VWA1 causes recessive hereditary motor neuropathy. *Brain* **2021**, *144* (2), 584.
- (104) Niu, X.; Zhang, F.; Ping, L.; Wang, Y.; Zhang, B.; Wang, J.; Chen, X. Vwa1 Knockout in Zebrafish Causes Abnormal Craniofacial Chondrogenesis by Regulating FGF Pathway. *Genes* **2023**, *14* (4), 838.
- (105) Sun, C.; Tian, X.; Jia, Y.; Yang, M.; Li, Y.; Fernig, D. G. Functions of Exogenous FGF Signals in Regulation of Fibroblast to Myofibroblast Differentiation and Extracellular Matrix Protein Expression. *Open Biol.* **2022**, *12* (9), 210356.
- (106) Shimbori, C.; Bellay, P. S.; Xia, J.; Gaudie, J.; Ask, K.; Ramos, C.; Becerril, C.; Pardo, A.; Selman, M.; Kolb, M. Fibroblast Growth Factor-1 Attenuates TGF- β 1-Induced Lung Fibrosis. *J. Pathol.* **2016**, *240* (2), 197–210.
- (107) Svystonyuk, D. A.; Ngu, J. M. C.; Mewhort, H. E. M.; Lipon, B. D.; Teng, G.; Guzzardi, D. G.; Malik, G.; Belke, D. D.; Fedak, P. W. M. Fibroblast Growth Factor-2 Regulates Human Cardiac Myofibroblast-Mediated Extracellular Matrix Remodeling. *J. Transl. Med.* **2015**, *13* (1), 1–11.
- (108) Koo, H. Y.; El-Baz, L. M. F.; House, S. L.; Cilvik, S. N.; Dorry, S. J.; Shoukry, N. M.; Salem, M. L.; Hafez, H. S.; Dulin, N. O.; Ornitz, D. M.; Guzy, R. D. Fibroblast Growth Factor 2 Decreases Bleomycin-Induced Pulmonary Fibrosis and Inhibits Fibroblast Collagen Production and Myofibroblast Differentiation. *J. Pathol.* **2018**, *246* (1), 54–66.
- (109) Rodríguez-Rivera, C.; Garcia, M. M.; Molina-Álvarez, M.; González-Martín, C.; Goicoechea, C. Clusterin Always Protecting. Synthesis, Function and Potential Issues. *Biomed. Pharmacother.* **2021**, *134*, 111174.
- (110) Peix, L.; Evans, I. C.; Pearce, D. R.; Simpson, J. K.; Maher, T. M.; McNulty, R. J. Diverse Functions of Clusterin Promote and Protect against the Development of Pulmonary Fibrosis. *Sci. Rep.* **2018**, *8* (1), 1906.
- (111) Seo, H. Y.; Lee, S. H.; Lee, J. H.; Kang, Y. N.; Choi, Y. K.; Hwang, J. S.; Park, K. G.; Jang, B. K.; Kim, M. K. Clusterin Attenuates Hepatic Fibrosis by Inhibiting Hepatic Stellate Cell Activation and Downregulating the Smad3 Signaling Pathway. *Cells* **2019**, *8* (11), 1442.
- (112) Jung, G. S.; Kim, M. K.; Jung, Y. A.; Kim, H. S.; Park, I. S.; Min, B. H.; Lee, K. U.; Kim, J. G.; Park, K. G.; Lee, I. K. Clusterin Attenuates the Development of Renal Fibrosis. *J. Am. Soc. Nephrol.* **2012**, *23* (1), 73–85.
- (113) Elbitar, S.; Renard, M.; Arnaud, P.; Hanna, N.; Jacob, M. P.; Guo, D. C.; Tsutsui, K.; Gross, M. S.; Kessler, K.; Tosolini, L.; Dattilo, V.; Dupont, S.; Jonquet, J.; Langeois, M.; Benarroch, L.; Aubart, M.; Ghaleb, Y.; Abou Khalil, Y.; Varret, M.; El Khoury, P.; Ho-Tin-Noé, B.; Alembik, Y.; Gaertner, S.; Isidor, B.; Gouya, L.; Milleron, O.; Sekiguchi, K.; Milewicz, D.; De Backer, J.; Le Goff, C.; Michel, J. B.; Jondeau, G.; Sakai, L. Y.; Boileau, C.; Abifadel, M. Pathogenic Variants in THSD4, Encoding the ADAMTS-like 6 Protein, Predispose to Inherited Thoracic Aortic Aneurysm. *Genet. Med.* **2021**, *23* (1), 111.
- (114) Rypdal, K. B.; Apte, S. S.; Lunde, I. G. Emerging Roles for the ADAMTS-like Family of Matricellular Proteins in Cardiovascular Disease through Regulation of the Extracellular Microenvironment. *Mol. Biol. Rep.* **2024**, *51* (1), 1–13.
- (115) Saito, M.; Kurokawa, M.; Oda, M.; Oshima, M.; Tsutsui, K.; Kosaka, K.; Nakao, K.; Ogawa, M.; Manabe, R. I.; Suda, N.; Ganjargal, G.; Hada, Y.; Noguchi, T.; Teranaka, T.; Sekiguchi, K.; Yoneda, T.; Tsuji, T. ADAMTS16 β Protein Rescues Fibrillin-1 Microfibril Disorder in a Marfan Syndrome Mouse Model through the

- Promotion of Fibrillin-1 Assembly. *J. Biol. Chem.* **2011**, *286* (44), 38602.
- (116) Rose, M.; Noetzel, E.; Kistermann, J.; Eschenbruch, J.; Rushrush, S.; Gan, L.; Knüchel, R.; Gaisa, N. T.; Dahl, E. The Ecm Modulator Itih5 Affects Cell Adhesion, Motility and Chemotherapeutic Response of Basal/Squamous-like (Basq) Bladder Cancer Cells. *Cells* **2021**, *10* (5), 1038.
- (117) Huwiler, A.; Pfeilschifter, J. Sphingolipid Signaling in Renal Fibrosis. *Matrix Biol.* **2018**, *68–69*, 230–247.
- (118) Wang, E.; He, X.; Zeng, M. The Role of S1P and the Related Signaling Pathway in the Development of Tissue Fibrosis. *Front. Pharmacol.* **2019**, *9* (JAN), 1504.
- (119) Ma, H. Y.; Li, Q.; Wong, W. R.; N'Diaye, E. N.; Caplazi, P.; Bender, H.; Huang, Z.; Arlantino, A.; Jeet, S.; Wong, A.; et al. LOXL4, but Not LOXL2, Is the Critical Determinant of Pathological Collagen Cross-Linking and Fibrosis in the Lung. *Sci. Adv.* **2023**, *9* (21), No. eadf0133.
- (120) Hirani, R.; Hanssen, E.; Gibson, M. A. LTBP-2 Specifically Interacts with the Amino-Terminal Region of Fibrillin-1 and Competes with LTBP-1 for Binding to This Microfibrillar Protein. *Matrix Biol.* **2007**, *26* (4), 213–223.
- (121) Robertson, I. B.; Horiguchi, M.; Zilberberg, L.; Dabovic, B.; Hadjiolova, K.; Rifkin, D. B. Latent TGF- β -Binding Proteins. *Matrix Biol.* **2015**, *47*, 44–53.
- (122) Wang, M. Y.; Liu, W. J.; Wu, L. Y.; Wang, G.; Zhang, C. L.; Liu, J. The Research Progress in Transforming Growth Factor-B2. *Cells* **2023**, *12* (23), 2739.
- (123) Sideek, M. A.; Smith, J.; Menz, C.; Adams, J. R. J.; Cowin, A. J.; Gibson, M. A. A Central Bioactive Region of LTBP-2 Stimulates the Expression of TGF-B1 in Fibroblasts via Akt and P38 Signalling Pathways. *Int. J. Mol. Sci.* **2017**, *18* (10), 2114.
- (124) Morel, P. S.; Duvivier, V.; Bertin, F.; Provost, N.; Hammoutene, A.; Hubert, E. L.; Gonzalez, A.; Tupinon-Mathieu, I.; Paradis, V.; Delerive, P.; Avila, M. A. Procollagen C-Proteinase Enhancer-1 (PCPE-1) Deficiency in Mice Reduces Liver Fibrosis but Not NASH Progression. *PLoS One* **2022**, *17* (2), No. e0263828.
- (125) Kessler, E.; Hassoun, E. Procollagen C-Proteinase Enhancer 1 (PCPE-1) in Liver Fibrosis. *Methods Mol. Biol.* **2019**, *1944*, 189–201.
- (126) Lagoutte, P.; Bettler, E.; Vadon-Le Goff, S.; Moali, C. Procollagen C-Proteinase Enhancer-1 (PCPE-1), a Potential Biomarker and Therapeutic Target for Fibrosis. *Matrix Biol. Plus* **2021**, *11*, 100062.
- (127) Péterfi, Z.; Donkó, Á.; Orient, A.; Sum, A.; Prókai, Á.; Molnár, B.; Veréb, Z.; Rajnavölgyi, É.; Kovács, K. J.; Müller, V.; Szabó, A. J.; Geiszt, M. Peroxidase Is Secreted and Incorporated into the Extracellular Matrix of Myofibroblasts and Fibrotic Kidney. *Am. J. Pathol.* **2009**, *175* (2), 725.
- (128) Cheng, G.; Shi, R. Mammalian Peroxidase (PXDN): From Physiology to Pathology. *Free Radical Biol. Med.* **2022**, *182*, 100.
- (129) Ghosh, A. K.; Vaughan, D. E. PAI-1 in Tissue Fibrosis. *J. Cell. Physiol.* **2012**, *227* (2), 493.
- (130) Matsushita, M.; Yamamoto, T.; Nishioka, K. Plasminogen Activator Inhibitor-1 Is Elevated, but Not Essential, in the Development of Bleomycin-Induced Murine Scleroderma. *Clin. Exp. Immunol.* **2005**, *139* (3), 429–438.
- (131) Sun, T.; Huang, Z.; Liang, W. C.; Yin, J.; Lin, W. Y.; Wu, J.; Vernes, J. M.; Lutman, J.; Caplazi, P.; Jeet, S.; et al. TGF β 2 and TGF β 3 Isoforms Drive Fibrotic Disease Pathogenesis. *Sci. Transl. Med.* **2021**, *13* (605), No. eabe0407.
- (132) Maheshwari, A.; Kelly, D. R.; Nicola, T.; Ambalavanan, N.; Jain, S. K.; Murphyullrich, J.; Athar, M.; Shimamura, M.; Bhandari, V.; Aprahamian, C.; Dimmitt, R. A.; Serra, R.; Ohls, R. K. TGF-B2 Suppresses Macrophage Cytokine Production and Mucosal Inflammatory Responses in the Developing Intestine. *Gastroenterology* **2011**, *140* (1), 242.
- (133) Soldano, S.; Pizzorni, C.; Paolino, S.; Trombetta, A. C.; Montagna, P.; Brizzolara, R.; Ruaro, B.; Sulli, A.; Cutolo, M.; Zissel, G. Alternatively Activated (M2) Macrophage Phenotype Is Inducible by Endothelin-1 in Cultured Human Macrophages. *PLoS One* **2016**, *11* (11), No. e0166433.
- (134) Tian, K.; Du, G.; Wang, X.; Wu, X.; Li, L.; Liu, W.; Wu, R. MMP-9 Secreted by M2-Type Macrophages Promotes Wilms' Tumour Metastasis through the PI3K/AKT Pathway. *Mol. Biol. Rep.* **2022**, *49* (5), 3469–3480.
- (135) Wang, Y.; Jiao, L.; Qiang, C.; Chen, C.; Shen, Z.; Ding, F.; Lv, L.; Zhu, T.; Lu, Y.; Cui, X. The Role of Matrix Metalloproteinase 9 in Fibrosis Diseases and Its Molecular Mechanisms. *Biomed. Pharmacother.* **2024**, *171*, 116116.
- (136) Kui Tan, T.; Zheng, G.; Hsu, T. T.; Ra Lee, S.; Zhang, J.; Zhao, Y.; Tian, X.; Wang, Y.; Min Wang, Y.; Cao, Q.; Wang, Y.; Lee, V. W. S.; Wang, C.; Zheng, D.; Alexander, S. I.; Thompson, E.; Harris, D. C. H. Matrix Metalloproteinase-9 of Tubular and Macrophage Origin Contributes to the Pathogenesis of Renal Fibrosis via Macrophage Recruitment through Osteopontin Cleavage. *Lab. Invest.* **2013**, *93* (4), 434–449.
- (137) Ducharme, A.; Frantz, S.; Aikawa, M.; Rabkin, E.; Lindsey, M.; Rohde, L. E.; Schoen, F. J.; Kelly, R. A.; Werb, Z.; Libby, P.; Lee, R. T. Targeted Deletion of Matrix Metalloproteinase-9 Attenuates Left Ventricular Enlargement and Collagen Accumulation after Experimental Myocardial Infarction. *J. Clin. Invest.* **2000**, *106* (1), 55–62.
- (138) Wang, X.; Zhou, Y.; Tan, R.; Xiong, M.; He, W.; Fang, L.; Wen, P.; Jiang, L.; Yang, J. Mice Lacking the Matrix Metalloproteinase-9 Gene Reduce Renal Interstitial Fibrosis in Obstructive Nephropathy. *Am. J. Physiol. Renal Physiol.* **2010**, *299* (5), F973–F982.
- (139) Tan, T. K.; Zheng, G.; Hsu, T. T.; Wang, Y.; Lee, V. W. S.; Tian, X.; Wang, Y.; Cao, Q.; Wang, Y.; Harris, D. C. H. Macrophage Matrix Metalloproteinase-9 Mediates Epithelial-Mesenchymal Transition in Vitro in Murine Renal Tubular Cells. *Am. J. Pathol.* **2010**, *176* (3), 1256–1270.
- (140) Giannandrea, M.; Parks, W. C. Diverse Functions of Matrix Metalloproteinases during Fibrosis. *Dis. Model. Mech.* **2014**, *7* (2), 193.
- (141) Liu, W.; Xiong, S.; Du, J.; Song, Y.; Wang, T.; Zhang, Y.; Dong, C.; Huang, Z. S.; He, Q.; Yu, Z.; et al. Deciphering Key Foreign Body Reaction-Related Transcription Factors and Genes Through Transcriptome Analysis. *Front. Mol. Biosci.* **2022**, *9*, 1.